PART III

Homomorphisms and Factor Groups

Section 13 Homomorphisms
Section 14 Factor Groups
Section 15 Factor-Group Computations and Simple Groups
Section 16 ‡Group Action on a Set
Section 17 †Applications of G-Sets to Counting

SECTION 13 HOMOMORPHISMS

Structure-Relating Maps

Let G and G' be groups. We are interested in maps from G to G' that relate the group structure of G to the group structure of G'. Such a map often gives us information about one of the groups from known structural properties of the other. An isomorphism $\phi : G \to G'$, if one exists, is an example of such a structure-relating map. If we know all about the group G and know that ϕ is an isomorphism, we immediately know all about the group structure of G', for it is structurally just a copy of G. We now consider more general structure-relating maps, weakening the conditions from those of an isomorphism by no longer requiring that the maps be one to one and onto. You see, those conditions are the purely *set-theoretic portion* of our definition of an isomorphism, and have nothing to do with the binary operations of G and of G'. The binary operations are what give us the *algebra* which is the focus of our study in this text. We keep just the homomorphism property of an isomorphism related to the binary operations for the definition we now make.

13.1 Definition A map ϕ of a group G into a group G' is a **homomorphism** if the homomorphism property

$$\phi(ab) = \phi(a)\phi(b) \tag{1}$$

holds for all $a, b \in G$. ■

‡ Section 16 is a prerequisite only for Sections 17 and 36.

† Section 17 is not required for the remainder of the text.

Let us now examine the idea behind the requirement (1) for a homomorphism $\phi : G \to G'$. In Eq. (1), the product ab on the left-hand side takes place in G, while the product $\phi(a)\phi(b)$ on the right-hand side takes place in G'. Thus Eq. (1) gives a relation between these binary operations, and hence between the two group structures.

For any groups G and G', there is always at least one homomorphism $\phi : G \to G'$, namely the **trivial homomorphism** defined by $\phi(g) = e'$ for all $g \in G$, where e' is the identity in G'. Equation (1) then reduces to the true equation $e' = e'e'$. No information about the structure of G or G' can be gained from the other group using this trivial homomorphism. We give an example illustrating how a homomorphism ϕ mapping G onto G' may give structural information about G'.

13.2 Example Let $\phi : G \to G'$ be a group homomorphism of G *onto* G'. We claim that if G is abelian, then G' must be abelian. Let $a', b' \in G'$. We must show that $a'b' = b'a'$. Since ϕ is onto G', there exist $a, b \in G$ such that $\phi(a) = a'$ and $\phi(b) = b'$. Since G is abelian, we have $ab = ba$. Using property (1), we have $a'b' = \phi(a)\phi(b) = \phi(ab) = \phi(ba) = \phi(b)\phi(a) = b'a'$, so G' is indeed abelian? ▲

Example 13.16 will give an illustration showing how information about G' may give information about G via a homomorphism $\phi : G \to G'$. We now give examples of homomorphisms for specific groups.

13.3 Example Let S_n be the symmetric group on n letters, and let $\phi : S_n \to \mathbb{Z}_2$ be defined by

$$\phi(\sigma) = \begin{cases} 0 & \text{if } \sigma \text{ is an even permutation,} \\ 1 & \text{if } \sigma \text{ is an odd permutation.} \end{cases}$$

Show that ϕ is a homomorphism.

Solution We must show that $\phi(\sigma\mu) = \phi(\sigma) + \phi(\mu)$ for all choices of $\sigma, \mu \in S_n$. Note that the operation on the right-hand side of this equation is written additively since it takes place in the group $\mathbb{Z}_2$. Verifying this equation amounts to checking just four cases:

σ odd and μ odd,

σ odd and μ even,

σ even and μ odd,

σ even and μ even.

Checking the first case, if σ and μ can both be written as a product of an odd number of transpositions, then $\sigma\mu$ can be written as the product of an even number of transpositions. Thus $\phi(\sigma\mu) = 0$ and $\phi(\sigma) + \phi(\mu) = 1 + 1 = 0$ in $\mathbb{Z}_2$. The other cases can be checked similarly. ▲

13.4 Example **(Evaluation Homomorphism)** Let F be the additive group of all functions mapping $\mathbb{R}$ into $\mathbb{R}$, let $\mathbb{R}$ be the additive group of real numbers, and let c be any real number. Let $\phi_c : F \to \mathbb{R}$ be the **evaluation homomorphism** defined by $\phi_c(f) = f(c)$ for $f \in F$. Recall that, by definition, the sum of two functions f and g is the function $f + g$ whose value at x is $f(x) + g(x)$. Thus we have

$$\phi_c(f + g) = (f + g)(c) = f(c) + g(c) = \phi_c(f) + \phi_c(g),$$

and Eq. (1) is satisfied, so we have a homomorphism. ▲

13.5 Example Let $\mathbb{R}^n$ be the additive group of column vectors with n real-number components. (This group is of course isomorphic to the direct product of $\mathbb{R}$ under addition with itself for n factors.) Let A be an $m \times n$ matrix of real numbers. Let $\phi : \mathbb{R}^n \to \mathbb{R}^m$ be defined by $\phi(\mathbf{v}) = A\mathbf{v}$ for each column vector $\mathbf{v} \in \mathbb{R}^n$. Then ϕ is a homomorphism, since for $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$, matrix algebra shows that $\phi(\mathbf{v} + \mathbf{w}) = A(\mathbf{v} + \mathbf{w}) = A\mathbf{v} + A\mathbf{w} = \phi(\mathbf{v}) + \phi(\mathbf{w})$. In linear algebra, such a map computed by multiplying a column vector on the left by a matrix A is known as a **linear transformation**. ▲

13.6 Example Let $GL(n, \mathbb{R})$ be the multiplicative group of all invertible $n \times n$ matrices. Recall that a matrix A is invertible if and only if its determinant, $\det(A)$, is nonzero. Recall also that for matrices $A, B \in GL(n, \mathbb{R})$ we have

$$\det(AB) = \det(A)\det(B).$$

This means that det is a homomorphism mapping $GL(n, \mathbb{R})$ into the multiplicative group $\mathbb{R}^*$ of nonzero real numbers. ▲

Homomorphisms of a group G into itself are often useful for studying the structure of G. Our next example gives a nontrivial homomorphism of a group into itself.

13.7 Example Let $r \in \mathbb{Z}$ and let $\phi_r : \mathbb{Z} \to \mathbb{Z}$ be defined by $\phi_r(n) = rn$ for all $n \in \mathbb{Z}$. For all $m, n \in \mathbb{Z}$, we have $\phi_r(m + n) = r(m + n) = rm + rn = \phi_r(m) + \phi_r(n)$ so ϕ_r is a homomorphism. Note that ϕ_0 is the trivial homomorphism, ϕ_1 is the identity map, and ϕ_{-1} maps $\mathbb{Z}$ *onto* $\mathbb{Z}$. For all other r in $\mathbb{Z}$, the map ϕ_r is not onto $\mathbb{Z}$. ▲

13.8 Example Let $G = G_1 \times G_2 \times \cdots \times G_i \times \cdots \times G_n$ be a direct product of groups. The **projection map** $\pi_i : G \to G_i$ where $\pi_i(g_1, g_2, \cdots, g_i, \cdots, g_n) = g_i$ is a homomorphism for each $i = 1, 2, \cdots, n$. This follows immediately from the fact that the binary operation of G coincides in the ith component with the binary operation in G_i. ▲

13.9 Example Let F be the additive group of continuous functions with domain $[0, 1]$ and let $\mathbb{R}$ be the additive group of real numbers. The map $\sigma : F \to \mathbb{R}$ defined by $\sigma(f) = \int_0^1 f(x)dx$ for $f \in F$ is a homomorphism, for

$$\begin{aligned}\sigma(f + g) &= \int_0^1 (f + g)(x)dx = \int_0^1 [f(x) + g(x)]dx \\ &= \int_0^1 f(x)dx + \int_0^1 g(x)dx = \sigma(f) + \sigma(g)\end{aligned}$$

for all $f, g \in F$. ▲

13.10 Example **(Reduction Modulo n)** Let γ be the natural map of $\mathbb{Z}$ into $\mathbb{Z}_n$ given by $\gamma(m) = r$, where r is the remainder given by the division algorithm when m is divided by n. Show that γ is a homomorphism.

Solution We need to show that

$$\gamma(s + t) = \gamma(s) + \gamma(t)$$

for $s, t \in \mathbb{Z}$. Using the division algorithm, we let

$$s = q_1 n + r_1 \tag{2}$$

and

$$t = q_2 n + r_2 \tag{3}$$

where $0 \leq r_i < n$ for $i = 1, 2$. If

$$r_1 + r_2 = q_3 n + r_3 \tag{4}$$

for $0 \leq r_3 < n$, then adding Eqs. (2) and (3) we see that

$$s + t = (q_1 + q_2 + q_3)n + r_3,$$

so that $\gamma(s + t) = r_3$.

From Eqs. (2) and (3) we see that $\gamma(s) = r_1$ and $\gamma(t) = r_2$. Equation (4) shows that the sum $r_1 + r_2$ in $\mathbb{Z}_n$ is equal to r_3 also.

Consequently $\gamma(s + t) = \gamma(s) + \gamma(t)$, so we do indeed have a homomorphism. ▲

Each of the homomorphisms in the preceding three examples is a many-to-one map. That is, different points of the domain of the map may be carried into the same point. Consider, for illustration, the homomorphism $\pi_1 : \mathbb{Z}_2 \times \mathbb{Z}_4 \to \mathbb{Z}_2$ in Example 13.8 We have

$$\pi_1(0, 0) = \pi_1(0, 1) = \pi_1(0, 2) = \pi_1(0, 3) = 0,$$

so four elements in $\mathbb{Z}_2 \times \mathbb{Z}_4$ are mapped into 0 in $\mathbb{Z}_2$ by π_1.

Composition of group homomorphisms is again a group homomorphism. That is, if $\phi : G \to G'$ and $\gamma : G' \to G''$ are both group homomorphisms then their composition $(\gamma \circ \phi) : G \to G''$, where $(\gamma \circ \phi)(g) = \gamma(\phi(g))$ for $g \in G$, is also a homomorphism. (See Exercise 49.)

Properties of Homomorphisms

We turn to some structural features of G and G' that are *preserved* by a homomorphism $\phi : G \to G'$. First we review set-theoretic definitions. Note the use of *square brackets* when we apply a function to a *subset* of its domain.

13.11 Definition Let ϕ be a mapping of a set X into a set Y, and let $A \subseteq X$ and $B \subseteq Y$. The **image** $\phi[A]$ **of** A **in** Y **under** ϕ is $\{\phi(a) \mid a \in A\}$. The set $\phi[X]$ is the **range of** ϕ. The **inverse image** $\phi^{-1}[B]$ **of** B **in** X is $\{x \in X \mid \phi(x) \in B\}$. ■

The first three properties of a homomorphism stated in the theorem that follows have already been encountered for the special case of an isomorphism; namely, in Theorem 3.14, Exercise 28 of Section 4, and Exercise 41 of Section 5. There they were really obvious because the structures of G and G' were identical. We will now see that they hold for structure-relating maps of groups, even if the maps are not one to one and onto. We do not consider them obvious in this new context.

13.12 Theorem Let ϕ be a homomorphism of a group G into a group G'.

1. If e is the identity element in G, then $\phi(e)$ is the identity element e' in G'.
2. If $a \in G$, then $\phi(a^{-1}) = \phi(a)^{-1}$.

3. If H is a subgroup of G, then $\phi[H]$ is a subgroup of G'.
4. If K' is a subgroup of G', then $\phi^{-1}[K']$ is a subgroup of G.

Loosely speaking, ϕ preserves the identity element, inverses, and subgroups.

Proof Let ϕ be a homomorphism of G into G'. Then

$$\phi(a) = \phi(ae) = \phi(a)\phi(e).$$

Multiplying on the left by $\phi(a)^{-1}$, we see that $e' = \phi(e)$. Thus $\phi(e)$ must be the identity element e' in G'. The equation

$$e' = \phi(e) = \phi(aa^{-1}) = \phi(a)\phi(a^{-1})$$

shows that $\phi(a^{-1}) = \phi(a)^{-1}$.

Turning to Statement (3), let H be a subgroup of G, and let $\phi(a)$ and $\phi(b)$ be any two elements in $\phi[H]$. Then $\phi(a)\phi(b) = \phi(ab)$, so we see that $\phi(a)\phi(b) \in \phi[H]$; thus, $\phi[H]$ is closed under the operation of G'. The fact that $e' = \phi(e)$ and $\phi(a^{-1}) = \phi(a)^{-1}$ completes the proof that $\phi[H]$ is a subgroup of G'.

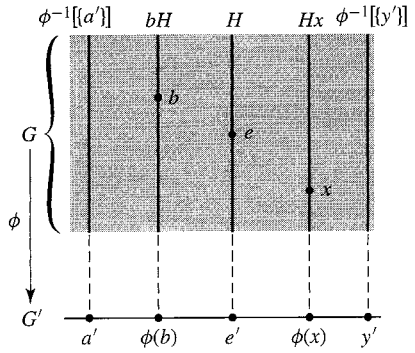
Going the other way for Statement (4), let K' be a subgroup of G'. Suppose a and b are in $\phi^{-1}[K']$. Then $\phi(a)\phi(b) \in K'$ since K' is a subgroup. The equation $\phi(ab) = \phi(a)\phi(b)$ shows that $ab \in \phi^{-1}[K']$. Thus $\phi^{-1}[K']$ is closed under the binary operation in G. Also, K' must contain the identity element $e' = \phi(e)$, so $e \in \phi^{-1}[K']$. If $a \in \phi^{-1}[K']$, then $\phi(a) \in K'$, so $\phi(a)^{-1} \in K'$. But $\phi(a)^{-1} = \phi(a^{-1})$, so we must have $a^{-1} \in \phi^{-1}[K']$. Hence $\phi^{-1}[K']$ is a subgroup of G. ◆

Let $\phi : G \to G'$ be a homomorphism and let e' be the identity element of G'. Now $\{e'\}$ is a subgroup of G', so $\phi^{-1}[\{e'\}]$ is a subgroup H of G by Statement (4) in Theorem 13.12. This subgroup is critical to the study of homomorphisms.

13.13 Definition Let $\phi : G \to G'$ be a homomorphism of groups. The subgroup $\phi^{-1}[\{e'\}] = \{x \in G \mid \phi(x) = e'\}$ is the **kernel of** ϕ, denoted by $\text{Ker}(\phi)$. ■

Example 13.5 discussed the homomorphism $\phi : \mathbb{R}^n \to \mathbb{R}^m$ given by $\phi(\mathbf{v}) = A\mathbf{v}$ where A is an $m \times n$ matrix. In this context, $\text{Ker}(\phi)$ is called the *null space* of A. It consists of all $\mathbf{v} \in \mathbb{R}^n$ such that $A\mathbf{v} = \mathbf{0}$, the zero vector.

Let $H = \text{Ker}(\phi)$ for a homomorphism $\phi : G \to G'$. We think of ϕ as "collapsing" H down onto e'. Theorem 13.15 that follows shows that for $g \in G$, the cosets gH and Hg are the same, and are collapsed onto the single element $\phi(g)$ by ϕ. That is $\phi^{-1}[\{\phi(g)\}] = gH = Hg$. (Be sure that you understand the reason for the uses of (), [], and $\{\}$ in $\phi^{-1}[\{\phi(g)\}]$.) We have attempted to symbolize this collapsing in Fig. 13.14, where the shaded rectangle represents G, the solid vertical line segments represent the cosets of $H = \text{Ker}(\phi)$, and the horizontal line at the bottom represents G'. We view ϕ as projecting the elements of G, which are in the shaded rectangle, straight down onto elements of G', which are on the horizontal line segment at the bottom. Notice the downward arrow labeled ϕ at the left, starting at G and ending at G'. Elements of $H = \text{Ker}(\phi)$ thus lie on the solid vertical line segment in the shaded box lying over e', as labeled at the top of the figure.



13.14 Figure Cosets of H collapsed by ϕ.

13.15 Theorem Let $\phi : G \to G'$ be a group homomorphism, and let $H = \text{Ker}(\phi)$. Let $a \in G$. Then the set

$$\phi^{-1}[\{\phi(a)\}] = \{x \in G \mid \phi(x) = \phi(a)\}$$

is the left coset aH of H, and is also the right coset Ha of H. Consequently, the two partitions of G into left cosets and into right cosets of H are the same.

Proof We want to show that

$$\{x \in G \mid \phi(x) = \phi(a)\} = aH.$$

> There is a standard way to show that two sets are equal; show that each is a subset of the other.

Suppose that $\phi(x) = \phi(a)$. Then

$$\phi(a)^{-1}\phi(x) = e',$$

where e' is the identity of G'. By Theorem 13.12, we know that $\phi(a)^{-1} = \phi(a^{-1})$, so we have

$$\phi(a^{-1})\phi(x) = e'.$$

Since ϕ is a homomorphism, we have

$$\phi(a^{-1})\phi(x) = \phi(a^{-1}x), \qquad \text{so} \qquad \phi(a^{-1}x) = e'.$$

But this shows that $a^{-1}x$ is in $H = \text{Ker}(\phi)$, so $a^{-1}x = h$ for some $h \in H$, and $x = ah \in aH$. This shows that

$$\{x \in G \mid \phi(x) = \phi(a)\} \subseteq aH.$$

To show containment in the other direction, let $y \in aH$, so that $y = ah$ for some $h \in H$. Then

$$\phi(y) = \phi(ah) = \phi(a)\phi(h) = \phi(a)e' = \phi(a),$$

so that $y \in \{x \in G \mid \phi(x) = \phi(a)\}$.

We leave the similar demonstration that $\{x \in G \mid \phi(x) = \phi(a)\} = Ha$ to Exercise 52. ◆

13.16 Example Equation 5 of Section 1 shows that $|z_1 z_2| = |z_1||z_2|$ for complex numbers z_1 and z_2. This means that the absolute value function $|\ |$ is a homomorphism of the group $\mathbb{C}^*$ of nonzero complex numbers under multiplication onto the group $\mathbb{R}^+$ of positive real numbers under multiplication. Since $\{1\}$ is a subgroup of $\mathbb{R}^+$, Theorem 13.12 shows again that the complex numbers of magnitude 1 form a subgroup U of $\mathbb{C}^*$. Recall that the complex numbers can be viewed as filling the coordinate plane, and that the magnitude of a complex number is its distance from the origin. Consequently, the cosets of U are circles with center at the origin. Each circle is collapsed by this homomorphism onto its point of intersection with the positive real axis. ▲

We give an illustration of Theorem 13.15 from calculus.

13.17 Example Let D be the additive group of all differentiable functions mapping $\mathbb{R}$ into $\mathbb{R}$, and let F be the additive group of all functions mapping $\mathbb{R}$ into $\mathbb{R}$. Then differentiation gives us a map $\phi : D \to F$, where $\phi(f) = f'$ for $f \in F$. We easily see that ϕ is a homomorphism, for $\phi(f + g) = (f + g)' = f' + g' = \phi(f) + \phi(g)$; the derivative of a sum is the sum of the derivatives.

Now $\text{Ker}(\phi)$ consists of all functions f such that $f' = 0$, the zero constant function. Thus $\text{Ker}(\phi)$ consists of all constant functions, which form a subgroup C of F. Let us find all functions in G mapped into x^2 by ϕ, that is, all functions whose derivative is x^2. Now we know that $x^3/3$ is one such function. By Theorem 13.15, all such functions form the coset $x^3/3 + C$. Doesn't this look familiar? ▲

We will often use the following corollary of Theorem 13.15.

13.18 Corollary A group homomorphism $\phi : G \to G'$ is a one-to-one map if and only if $\text{Ker}(\phi) = \{e\}$.

Proof If $\text{Ker}(\phi) = \{e\}$, then for every $a \in G$, the elements mapped into $\phi(a)$ are precisely the elements of the left coset $a\{e\} = \{a\}$, which shows that ϕ is one to one.

Conversely, suppose ϕ is one to one. Now by Theorem 13.12, we know that $\phi(e) = e'$, the identity element of G'. Since ϕ is one to one, we see that e is the only element mapped into e' by ϕ, so $\text{Ker}(\phi) = \{e\}$. ◆

In view of Corollary 13.18, we modify the outline given prior to Example 3.8 for showing that a map ϕ is an isomorphism of binary structures when the structures are groups G and G'.

To Show $\phi : G \to G'$ Is an Isomorphism

Step 1 Show ϕ is a homomorphism.

Step 2 Show $\text{Ker}(\phi) = \{e\}$.

Step 3 Show ϕ maps G onto G'.

Theorem 13.15 shows that the kernel of a group homomorphism $\phi : G \to G'$ is a subgroup H of G whose left and right cosets coincide, so that $gH = Hg$ for all $g \in G$. We will see in Section 14 that when left and right cosets coincide, we can form a coset group, as discussed intuitively in Section 10. Furthermore, we will see that H then appears as the kernel of a homomorphism of G onto this coset group in a very natural way. Such subgroups H whose left and right cosets coincide are very useful in studying a group, and are given a special name. We will work with them a lot in Section 14.

■ Historical Note

Normal subgroups were introduced by Evariste Galois in 1831 as a tool for deciding whether a given polynomial equation was solvable by radicals. Galois noted that a subgroup H of a group G of permutations induced two decompositions of G into what we call *left cosets* and *right cosets*. If the two decompositions coincide, that is, if the left cosets are the same as the right cosets, Galois called the decomposition *proper*. Thus a subgroup giving a proper decomposition is what we call a *normal subgroup*. Galois stated that if the group of permutations of the roots of an equation has a proper decomposition, then one can solve the given equation if one can first solve an equation corresponding to the subgroup H and then an equation corresponding to the cosets.

Camille Jordan, in his commentaries on Galois's work in 1865 and 1869, elaborated on these ideas considerably. He also defined normal subgroups, although without using the term, essentially as on this page and likewise gave the first definition of a simple group (page 149).

13.19 Definition A subgroup H of a group G is **normal** if its left and right cosets coincide, that is, if $gH = Hg$ for all $g \in G$. ■

Note that all subgroups of abelian groups are normal.

13.20 Corollary If $\phi : G \to G'$ is a group homomorphism, then $\text{Ker}(\phi)$ is a normal subgroup of G.

Proof This follows immediately from the last sentence in the statement of Theorem 13.15 and Definition 13.19. ◆

For any group homomorphism $\phi : G \to G'$, two things are of primary importance: the *kernel of* ϕ, and the *image* $\phi[G]$ *of* G *in* G'. We have indicated the importance of

Ker(ϕ). Section 14 will indicate the importance of the image $\phi[G]$. Exercise 44 asks us to show that if $|G|$ is finite, then $|\phi[G]|$ is finite and is a divisor of $|G|$.

EXERCISES 13

Computations

In Exercises 1 through 15, determine whether the given map ϕ is a homomorphism. [*Hint:* The straightforward way to proceed is to check whether $\phi(ab) = \phi(a)\phi(b)$ for all a and b in the domain of ϕ. However, if we should happen to notice that $\phi^{-1}[\{e'\}]$ is not a subgroup whose left and right cosets coincide, or that ϕ does not satisfy the properties given in Exercise 44 or 45 for finite groups, then we can say at once that ϕ is not a homomorphism.]

1. Let $\phi : \mathbb{Z} \to \mathbb{R}$ under addition be given by $\phi(n) = n$.
2. Let $\phi : \mathbb{R} \to \mathbb{Z}$ under addition be given by $\phi(x) =$ the greatest integer $\leq x$.
3. Let $\phi : \mathbb{R}^* \to \mathbb{R}^*$ under multiplication be given by $\phi(x) = |x|$.
4. Let $\phi : \mathbb{Z}_6 \to \mathbb{Z}_2$ be given by $\phi(x) =$ the remainder of x when divided by 2, as in the division algorithm.
5. Let $\phi : \mathbb{Z}_9 \to \mathbb{Z}_2$ be given by $\phi(x) =$ the remainder of x when divided by 2, as in the division algorithm.
6. Let $\phi : \mathbb{R} \to \mathbb{R}^*$, where $\mathbb{R}$ is additive and $\mathbb{R}^*$ is multiplicative, be given by $\phi(x) = 2^x$.
7. Let $\phi_i : G_i \to G_1 \times G_2 \times \cdots \times G_i \times \cdots \times G_r$ be given by $\phi_i(g_i) = (e_1, e_2, \ldots, g_i, \ldots, e_r)$, where $g_i \in G_i$ and e_j is the identity element of G_j. This is an **injection map**. Compare with Example 13.8.
8. Let G be any group and let $\phi : G \to G$ be given by $\phi(g) = g^{-1}$ for $g \in G$.
9. Let F be the additive group of functions mapping $\mathbb{R}$ into $\mathbb{R}$ having derivatives of all orders. Let $\phi : F \to F$ be given by $\phi(f) = f''$, the second derivative of f.
10. Let F be the additive group of all continuous functions mapping $\mathbb{R}$ into $\mathbb{R}$. Let $\mathbb{R}$ be the additive group of real numbers, and let $\phi : F \to \mathbb{R}$ be given by
$$\phi(f) = \int_0^4 f(x)dx.$$
11. Let F be the additive group of all functions mapping $\mathbb{R}$ into $\mathbb{R}$, and let $\phi : F \to F$ be given by $\phi(f) = 3f$.
12. Let M_n be the additive group of all $n \times n$ matrices with real entries, and let $\mathbb{R}$ be the additive group of real numbers. Let $\phi(A) = \det(A)$, the determinant of A, for $A \in M_n$.
13. Let M_n and $\mathbb{R}$ be as in Exercise 12. Let $\phi(A) = \text{tr}(A)$ for $A \in M_n$, where the **trace** tr(A) is the sum of the elements on the main diagonal of A, from the upper-left to the lower-right corner.
14. Let $GL(n, \mathbb{R})$ be the multiplicative group of invertible $n \times n$ matrices, and let $\mathbb{R}$ be the additive group of real numbers. Let $\phi : GL(n, \mathbb{R}) \to \mathbb{R}$ be given by $\phi(A) = \text{tr}(A)$, where tr(A) is defined in Exercise 13.
15. Let F be the multiplicative group of all continuous functions mapping $\mathbb{R}$ into $\mathbb{R}$ that are nonzero at every $x \in \mathbb{R}$. Let $\mathbb{R}^*$ be the multiplicative group of nonzero real numbers. Let $\phi : F \to \mathbb{R}^*$ be given by $\phi(f) = \int_0^1 f(x)dx$.

In Exercises 16 through 24, compute the indicated quantities for the given homomorphism ϕ. (See Exercise 46.)

16. Ker(ϕ) for $\phi : S_3 \to \mathbb{Z}_2$ in Example 13.3
17. Ker(ϕ) and $\phi(25)$ for $\phi : \mathbb{Z} \to \mathbb{Z}_7$ such that $\phi(1) = 4$
18. Ker(ϕ) and $\phi(18)$ for $\phi : \mathbb{Z} \to \mathbb{Z}_{10}$ such that $\phi(1) = 6$

19. Ker(ϕ) and $\phi(20)$ for $\phi : \mathbb{Z} \to S_8$ such that $\phi(1) = (1, 4, 2, 6)(2, 5, 7)$

20. Ker(ϕ) and $\phi(3)$ for $\phi : \mathbb{Z}_{10} \to \mathbb{Z}_{20}$ such that $\phi(1) = 8$

21. Ker(ϕ) and $\phi(14)$ for $\phi : \mathbb{Z}_{24} \to S_8$ where $\phi(1) = (2, 5)(1, 4, 6, 7)$

22. Ker(ϕ) and $\phi(-3, 2)$ for $\phi : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ where $\phi(1, 0) = 3$ and $\phi(0, 1) = -5$

23. Ker(ϕ) and $\phi(4, 6)$ for $\phi : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z} \times \mathbb{Z}$ where $\phi(1, 0) = (2, -3)$ and $\phi(0, 1) = (-1, 5)$

24. Ker(ϕ) and $\phi(3, 10)$ for $\phi : \mathbb{Z} \times \mathbb{Z} \to S_{10}$ where $\phi(1, 0) = (3, 5)(2, 4)$ and $\phi(0, 1) = (1, 7)(6, 10, 8, 9)$

25. How many homomorphisms are there of $\mathbb{Z}$ onto $\mathbb{Z}$?

26. How many homomorphisms are there of $\mathbb{Z}$ into $\mathbb{Z}$?

27. How many homomorphisms are there of $\mathbb{Z}$ into $\mathbb{Z}_2$?

28. Let G be a group, and let $g \in G$. Let $\phi_g : G \to G$ be defined by $\phi_g(x) = gx$ for $x \in G$. For which $g \in G$ is ϕ_g a homomorphism?

29. Let G be a group, and let $g \in G$. Let $\phi_g : G \to G$ be defined by $\phi_g(x) = gxg^{-1}$ for $x \in G$. For which $g \in G$ is ϕ_g a homomorphism?

Concepts

In Exercises 30 and 31, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

30. A *homomorphism* is a map such that $\phi(xy) = \phi(x)\phi(y)$.

31. Let $\phi : G \to G'$ be a homomorphism of groups. The *kernel of* ϕ is $\{x \in G \mid \phi(x) = e'\}$ where e' is the identity in G'.

32. Mark each of the following true or false.

_____ **a.** A_n is a normal subgroup of S_n.

_____ **b.** For any two groups G and G', there exists a homomorphism of G into G'.

_____ **c.** Every homomorphism is a one-to-one map.

_____ **d.** A homomorphism is one to one if and only if the kernel consists of the identity element alone.

_____ **e.** The image of a group of 6 elements under some homomorphism may have 4 elements. (See Exercise 44.)

_____ **f.** The image of a group of 6 elements under a homomorphism may have 12 elements.

_____ **g.** There is a homomorphism of some group of 6 elements into some group of 12 elements.

_____ **h.** There is a homomorphism of some groups of 6 elements into some group of 10 elements.

_____ **i.** A homomorphism may have an empty kernel.

_____ **j.** It is not possible to have a nontrivial homomorphism of some finite group into some infinite group.

In Exercises 33 through 43, give an example of a nontrivial homomorphism ϕ for the given groups, if an example exists. If no such homomorphism exists, explain why that is so. You may use Exercises 44 and 45.

33. $\phi : \mathbb{Z}_{12} \to \mathbb{Z}_5$

34. $\phi : \mathbb{Z}_{12} \to \mathbb{Z}_4$

35. $\phi : \mathbb{Z}_2 \times \mathbb{Z}_4 \to \mathbb{Z}_2 \times \mathbb{Z}_5$

36. $\phi : \mathbb{Z}_3 \to \mathbb{Z}$

37. $\phi : \mathbb{Z}_3 \to S_3$

38. $\phi : \mathbb{Z} \to S_3$

39. $\phi : \mathbb{Z} \times \mathbb{Z} \to 2\mathbb{Z}$

40. $\phi : 2\mathbb{Z} \to \mathbb{Z} \times \mathbb{Z}$

41. $\phi : D_4 \to S_3$

42. $\phi : S_3 \to S_4$

43. $\phi : S_4 \to S_3$

Theory

44. Let $\phi : G \to G'$ be a group homomorphism. Show that if $|G|$ is finite, then $|\phi[G]|$ is finite and is a divisor of $|G|$.

45. Let $\phi : G \to G'$ be a group homomorphism. Show that if $|G'|$ is finite, then, $|\phi[G]|$ is finite and is a divisor of $|G'|$.

46. Let a group G be generated by $\{a_i \mid i \in I\}$, where I is some indexing set and $a_i \in G$ for all $i \in I$. Let $\phi : G \to G'$ and $\mu : G \to G'$ be two homomorphisms from G into a group G', such that $\phi(a_i) = \mu(a_i)$ for every $i \in I$. Prove that $\phi = \mu$. [Thus, for example, a homomorphism of a cyclic group is completely determined by its value on a generator of the group.] [*Hint*: Use Theorem 7.6 and, of course, Definition 13.1.]

47. Show that any group homomorphism $\phi : G \to G'$ where $|G|$ is a prime must either be the trivial homomorphism or a one-to-one map.

48. The **sign of an even permutation** is $+1$ and the **sign of an odd permutation** is -1. Observe that the map $\text{sgn}_n : S_n \to \{1, -1\}$ defined by

$$\text{sgn}_n(\sigma) = \text{sign of } \sigma$$

is a homomorphism of S_n onto the multiplicative group $\{1, -1\}$. What is the kernel? Compare with Example 13.3.

49. Show that if G, G', and G'' are groups and if $\phi : G \to G'$ and $\gamma : G' \to G''$ are homomorphisms, then the composite map $\gamma\phi : G \to G''$ is a homomorphism.

50. Let $\phi : G \to H$ be a group homomorphism. Show that $\phi[G]$ is abelian if and only if for all $x, y \in G$, we have $xyx^{-1}y^{-1} \in \text{Ker}(\phi)$.

51. Let G be any group and let a be any element of G. Let $\phi : \mathbb{Z} \to G$ be defined by $\phi(n) = a^n$. Show that ϕ is a homomorphism. Describe the image and the possibilities for the kernel of ϕ.

52. Let $\phi : G \to G'$ be a homomorphism with kernel H and let $a \in G$. Prove the set equality $\{x \in G \mid \phi(x) = \phi(a)\} = Ha$.

53. Let G be a group, Let $h, k \in G$ and let $\phi : \mathbb{Z} \times \mathbb{Z} \to G$ be defined by $\phi(m, n) = h^m k^n$. Give a necessary and sufficient condition, involving h and k, for ϕ to be a homomorphism. Prove your condition.

54. Find a necessary and sufficient condition on G such that the map ϕ described in the preceding exercise is a homomorphism for *all* choices of $h, k \in G$.

55. Let G be a group, h an element of G, and n a positive integer. Let $\phi : \mathbb{Z}_n \to G$ be defined by $\phi(i) = h^i$ for $0 \leq i \leq n$. Give a necessary and sufficient condition (in terms of h and n) for ϕ to be a homomorphism. Prove your assertion.

SECTION 14 FACTOR GROUPS

Let H be a subgroup of a finite group G. Suppose we write a table for the group operation of G, listing element heads at the top and at the left as they occur in the left cosets of H. We illustrated this in Section 10. The body of the table may break up into blocks corresponding to the cosets (Table 10.5), giving a group operation on the cosets, or they may not break up that way (Table 10.9). We start this section by showing that if H is the kernel of a group homomorphism $\phi : G \to G'$, then the cosets of H (remember that left and right cosets then coincide) are indeed elements of a group whose binary operation is derived from the group operation of G.

Factor Groups from Homomorphisms

Let G be a group and let S be a set having the same cardinality as G. Then there is a one-to-one correspondence $\leftrightarrow$ between S and G. We can use $\leftrightarrow$ to define a binary operation on S, making S into a group isomorphic to G. Naively, we simply use the correspondence to rename each element of G by the name of its corresponding (under $\leftrightarrow$) element in S. We can describe explicitly the computation of xy for $x, y \in S$ as follows:

$$\text{if } x \leftrightarrow g_1 \quad \text{and} \quad y \leftrightarrow g_2 \quad \text{and} \quad z \leftrightarrow g_1 g_2, \quad \text{then} \quad xy = z. \tag{1}$$

The direction $\rightarrow$ of the one-to-one correspondence $s \leftrightarrow g$ between $s \in S$ and $g \in G$ gives us a one-to-one function μ mapping S onto G. (Of course, the direction $\leftarrow$ of $\leftrightarrow$ gives us the inverse function μ^{-1}). Expressed in terms of μ, the computation (1) of xy for $x, y \in S$ becomes

$$\text{if } \mu(x) = g_1 \quad \text{and} \quad \mu(y) = g_2 \quad \text{and} \quad \mu(z) = g_1 g_2, \quad \text{then} \quad xy = z. \tag{2}$$

The map $\mu : S \rightarrow G$ now becomes an isomorphism mapping the group S onto the group G. Notice that from (2), we obtain $\mu(xy) = \mu(z) = g_1 g_2 = \mu(x)\mu(y)$, the required homomorphism property.

Let G and G' be groups, let $\phi : G \rightarrow G'$ be a homomorphism, and let $H = \text{Ker}(\phi)$. Theorem 13.15 shows that for $a \in G$, we have $\phi^{-1}[\{\phi(a)\}] = aH = Ha$. We have a one-to-one correspondence $aH \leftrightarrow \phi(a)$ between cosets of H in G and elements of the subgroup $\phi[G]$ of G'. Remember that if $x \in aH$, so that $x = ah$ for some $h \in H$, then $\phi(x) = \phi(ah) = \phi(a)\phi(h) = \phi(a)e' = \phi(a)$, so the computation of the element of $\phi[G]$ corresponding to the coset $aH = xH$ is the same whether we compute it as $\phi(a)$ or as $\phi(x)$. Let us denote the set of all cosets of H by G/H. (We read G/H as "G over H" or as "G modulo H" or as "G mod H," but *never* as "G divided by H.")

In the preceding paragraph, we started with a homomorphism $\phi : G \rightarrow G'$ having kernel H, and we finished with the set G/H of cosets in one-to-one correspondence with the elements of the group $\phi[G]$. In our work above that, we had a set S with elements in one-to-one correspondence with a those of a group G, and we made S into a group isomorphic to G with an isomorphism μ. Replacing S by G/H and replacing G by $\phi[G]$ in that construction, we can consider G/H to be a group isomorphic to $\phi[G]$ with that isomorphism μ. In terms of G/H and $\phi[G]$, the computation (2) of the product $(xH)(yH)$ for $xH, yH \in G/H$ becomes

$$\begin{aligned} &\text{if } \mu(xH) = \phi(x) \quad \text{and} \quad \mu(yH) = \phi(y) \quad \text{and} \quad \mu(zH) = \phi(x)\phi(y), \\ &\text{then} \quad (xH)(yH) = zH. \end{aligned} \tag{3}$$

But because ϕ is a homomorphism, we can easily find $z \in G$ such that $\mu(zH) = \phi(x)\phi(y)$; namely, we take $z = xy$ in G, and find that

$$\mu(zH) = \mu(xyH) = \phi(xy) = \phi(x)\phi(y).$$

This shows that the product $(xH)(yH)$ of two cosets is the coset $(xy)H$ that contains the product xy of x and y in G. While this computation of $(xH)(yH)$ may seem to depend on our choices x from xH and y from yH, our work above shows it does not. We demonstrate it again here because it is such an important point. If $h_1, h_2 \in H$ so that xh_1 is an element of xH and yh_2 is an element of yH, then there exists $h_3 \in H$ such

that $h_1 y = y h_3$ because $Hy = yH$ by Theorem 13.15. Thus we have

$$(xh_1)(yh_2) = x(h_1 y)h_2 = x(yh_3)h_2 = (xy)(h_3 h_2) \in (xy)H,$$

so we obtain the same coset. Computation of the product of two cosets is accomplished by *choosing* an element from each coset and taking, as product of the cosets, the coset that contains the product in G of the choices. Any time we define something (like a product) in terms of choices, it is important to show that it is **well defined,** which means that it is independent of the choices made. This is precisely what we have just done. We summarize this work in a theorem.

14.1 Theorem Let $\phi : G \to G'$ be a group homomorphism with kernel H. Then the cosets of H form a **factor group**, G/H, where $(aH)(bH) = (ab)H$. Also, the map $\mu : G/H \to \phi[G]$ defined by $\mu(aH) = \phi(a)$ is an isomorphism. Both coset multiplication and μ are well defined, independent of the choices a and b from the cosets.

14.2 Example Example 13.10 considered the map $\gamma : \mathbb{Z} \to \mathbb{Z}_n$, where $\gamma(m)$ is the remainder when m is divided by n in accordance with the division algorithm. We know that γ is a homomorphism. Of course, $\text{Ker}(\gamma) = n\mathbb{Z}$. By Theorem 14.1, we see that the factor group $\mathbb{Z}/n\mathbb{Z}$ is isomorphic to $\mathbb{Z}_n$. The cosets of $n\mathbb{Z}$ are the *residue classes modulo n*. For example, taking $n = 5$, we see the cosets of $5\mathbb{Z}$ are

$$\begin{aligned}
5\mathbb{Z} &= \{\cdots, -10, -5, 0, 5, 10, \cdots\},\\
1 + 5\mathbb{Z} &= \{\cdots, -9, -4, 1, 6, 11, \cdots\},\\
2 + 5\mathbb{Z} &= \{\cdots, -8, -3, 2, 7, 12, \cdots\},\\
3 + 5\mathbb{Z} &= \{\cdots, -7, -2, 3, 8, 13, \cdots\},\\
4 + 5\mathbb{Z} &= \{\cdots, -6, -1, 4, 9, 14, \cdots\}.
\end{aligned}$$

Note that the isomorphism $\mu : \mathbb{Z}/5\mathbb{Z} \to \mathbb{Z}_5$ of Theorem 14.1 assigns to each coset of $5\mathbb{Z}$ its smallest nonnegative element. That is, $\mu(5\mathbb{Z}) = 0$, $\mu(1 + 5\mathbb{Z}) = 1$, etc. ▲

It is very important that we learn how to compute in a factor group. We can multiply (add) two cosets by choosing *any* two representative elements, multiplying (adding) them and finding the coset in which the resulting product (sum) lies.

14.3 Example Consider the factor group $\mathbb{Z}/5\mathbb{Z}$ with the cosets shown above. We can add $(2 + 5\mathbb{Z}) + (4 + 5\mathbb{Z})$ by choosing 2 and 4, finding $2 + 4 = 6$, and noticing that 6 is in the coset $1 + 5\mathbb{Z}$. We could equally well add these two cosets by choosing 27 in $2 + 5\mathbb{Z}$ and -16 in $4 + 5\mathbb{Z}$; the sum $27 + (-16) = 11$ is also in the coset $1 + 5\mathbb{Z}$. ▲

The factor groups $\mathbb{Z}/n\mathbb{Z}$ in the preceding example are classics. Recall that we refer to the cosets of $n\mathbb{Z}$ as *residue classes modulo n*. Two integers in the same coset are *congruent modulo n*. This terminology is carried over to other factor groups. A factor group G/H is often called the **factor group of G modulo H**. Elements in the same coset of H are often said to be **congruent modulo H**. By abuse of notation, we may sometimes write $\mathbb{Z}/n\mathbb{Z} = \mathbb{Z}_n$ and think of $\mathbb{Z}_n$ as the additive group of residue classes of $\mathbb{Z}$ modulo $\langle n \rangle$, or abusing notation further, modulo n.

Factor Groups from Normal Subgroups

So far, we have obtained factor groups only from homomorphisms. Let G be a group and let H be a subgroup of G. Now H has both left cosets and right cosets, and in general, a left coset aH need not be the same set as the right coset Ha. Suppose we try to define a binary operation on left cosets by defining

$$(aH)(bH) = (ab)H \tag{4}$$

as in the statement of Theorem 14.1 Equation 4 attempts to define left coset multiplication by choosing representatives a and b from the cosets. Equation 4 is meaningless unless it gives a *well-defined* operation, independent of the representative elements a and b chosen from the cosets. The theorem that follows shows that Eq. 4 gives a well-defined binary operation if and only if H is a normal subgroup of G.

14.4 Theorem Let H be a subgroup of a group G. Then left coset multiplication is well defined by the equation

$$(aH)(bH) = (ab)H$$

if and only if H is a normal subgroup of G.

Proof Suppose first that $(aH)(bH) = (ab)H$ does give a well-defined binary operation on left cosets. Let $a \in G$. We want to show that aH and Ha are the same set. We use the standard technique of showing that each is a subset of the other.

Let $x \in aH$. Choosing representatives $x \in aH$ and $a^{-1} \in a^{-1}H$, we have $(xH)(a^{-1}H) = (xa^{-1})H$. On the other hand, choosing representatives $a \in aH$ and $a^{-1} \in a^{-1}H$, we see that $(aH)(a^{-1}H) = eH = H$. Using our assumption that left coset multiplication by representatives is well defined, we must have $xa^{-1} = h \in H$. Then $x = ha$, so $x \in Ha$ and $aH \subseteq Ha$. We leave the symmetric proof that $Ha \subseteq aH$ to Exercise 25.

We turn now to the converse: If H is a normal subgroup, then left coset multiplication by representatives is well-defined. Due to our hypothesis, we can simply say *cosets,* omitting *left* and *right.* Suppose we wish to compute $(aH)(bH)$. Choosing $a \in aH$ and $b \in bH$, we obtain the coset $(ab)H$. Choosing different representatives $ah_1 \in aH$ and $bh_2 \in bH$, we obtain the coset ah_1bh_2H. We must show that these are the same coset. Now $h_1b \in Hb = bH$, so $h_1b = bh_3$ for some $h_3 \in H$. Thus

$$(ah_1)(bh_2) = a(h_1b)h_2 = a(bh_3)h_2 = (ab)(h_3h_2)$$

and $(ab)(h_3h_2) \in (ab)H$. Therefore, ah_1bh_2 is in $(ab)H$. ◆

Theorem 14.4 shows that if left and right cosets of H coincide, then Eq. 4 gives a well-defined binary operation on cosets. We wonder whether the cosets do form a group with such coset multiplication. This is indeed true.

14.5 Corollary Let H be a normal subgroup of G. Then the cosets of H form a group G/H under the binary operation $(aH)(bH) = (ab)H$. ▲

Proof Computing, $(aH)[(bH)(cH)] = (aH)[(bc)H] = [a(bc)]H$, and similarly, we have $[(aH)(bH)](cH) = [(ab)c]H$, so associativity in G/H follows from associativity in G. Because $(aH)(eH) = (ae)H = aH = (ea)H = (eH)(aH)$, we see that $eH = H$ is the identity element in G/H. Finally, $(a^{-1}H)(aH) = (a^{-1}a)H = eH = (aa^{-1})H = (aH)(a^{-1}H)$ shows that $a^{-1}H = (aH)^{-1}$. ◆

14.6 Definition The group G/H in the preceding corollary is the **factor group** (or **quotient group**) of G by H. ■

14.7 Example Since $\mathbb{Z}$ is an abelian group, $n\mathbb{Z}$ is a normal subgroup. Corollary 14.5 allows us to construct the factor group $\mathbb{Z}/n\mathbb{Z}$ with no reference to a homomorphism. As we observed in Example 14.2, $\mathbb{Z}/n\mathbb{Z}$ is isomorphic to $\mathbb{Z}_n$. ▲

14.8 Example Consider the abelian group $\mathbb{R}$ under addition, and let $c \in \mathbb{R}^+$. The cyclic subgroup $\langle c \rangle$ of $\mathbb{R}$ contains as elements

$$\cdots -3c, -2c, -c, 0, c, 2c, 3c, \cdots.$$

Every coset of $\langle c \rangle$ contains just one element of x such that $0 \leq x < c$. If we choose these elements as representatives of the cosets when computing in $\mathbb{R}/\langle c \rangle$, we find that we are computing their sum modulo c as discussed for the computation in $\mathbb{R}_c$ in Section 1. For example, if $c = 5.37$, then the sum of the cosets $4.65 + \langle 5.37 \rangle$ and $3.42 + \langle 5.37 \rangle$ is the coset $8.07 + \langle 5.37 \rangle$, which contains $8.07 - 5.37 = 2.7$, which is $4.65 +_{5.37} 3.42$. Working with these coset elements x where $0 \leq x < c$, we thus see that the group $\mathbb{R}_c$ of Example 4.2 is isomorphic to $\mathbb{R}/\langle c \rangle$ under an isomorphism ψ where $\psi(x) = x + \langle c \rangle$ for all $x \in \mathbb{R}_c$. Of course, $\mathbb{R}/\langle c \rangle$ is then also isomorphic to the circle group U of complex numbers of magnitude 1 under multiplication. ▲

We have seen that the group $\mathbb{Z}/\langle n \rangle$ is isomorphic to the group $\mathbb{Z}_n$, and as a set, $\mathbb{Z}_n = \{0, 1, 3, 4, \cdots, n-1\}$, the set of nonnegative integers less than n. Example 14.8 shows that the group $\mathbb{R}/\langle c \rangle$ is isomorphic to the group $\mathbb{R}_c$. In Section 1, we choose the notation $\mathbb{R}_c$ rather than the conventional $[0, c)$ for the half-open interval of nonnegative real numbers less than c. We did that to bring out now the comparison of these factor groups of $\mathbb{Z}$ with these factor groups of $\mathbb{R}$.

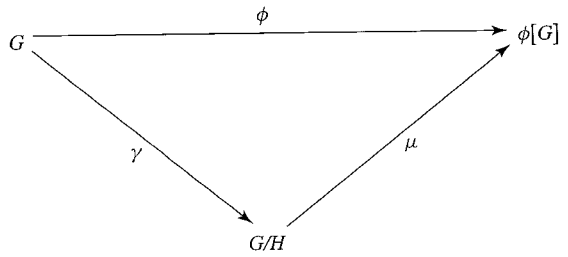
The Fundamental Homomorphism Theorem

We have seen that every homomorphism $\phi : G \to G'$ gives rise to a natural factor group (Theorem 14.1), namely, $G/\text{Ker}(\phi)$. We now show that each factor group G/H gives rise to a natural homomorphism having H as kernel.

14.9 Theorem Let H be a normal subgroup of G. Then $\gamma : G \to G/H$ given by $\gamma(x) = xH$ is a homomorphism with kernel H.

Proof Let $x, y \in G$. Then

$$\gamma(xy) = (xy)H = (xH)(yH) = \gamma(x)\gamma(y),$$



14.10 Figure

so γ is a homomorphism. Since $xH = H$ if an only if $x \in H$, we see that the kernel of γ is indeed H. ◆

We have seen in Theorem 14.1 that if $\phi : G \to G'$ is a homomorphism with kernel H, then $\mu : G/H \to \phi[G]$ where $\mu(gH) = \phi(g)$ is an isomorphism. Theorem 14.9 shows that $\gamma : G \to G/H$ defined by $\gamma(g) = gH$ is a homomorphism. Figure 14.10 shows these groups and maps. We see that the homomorphism ϕ can be *factored*, $\phi = \mu\gamma$, where γ is a homomorphism and μ is an isomorphism of G/H with $\phi[G]$. We state this as a theorem.

14.11 Theorem **(The Fundamental Homomorphism Theorem)** Let $\phi : G \to G'$ be a group homomorphism with kernel H. Then $\phi[G]$ is a group, and $\mu : G/H \to \phi[G]$ given by $\mu(gH) = \phi(g)$ is an isomorphism. If $\gamma : G \to G/H$ is the homomorphism given by $\gamma(g) = gH$, then $\phi(g) = \mu\gamma(g)$ for each $g \in G$.

The isomorphism μ in Theorem 14.11 is referred to as a *natural* or *canonical* isomorphism, and the same adjectives are used to describe the homomorphism γ. There may be other isomorphisms and homomorphisms for these same groups, but the maps μ and γ have a special status with ϕ and are uniquely determined by Theorem 14.11.

In summary, every homomorphism with domain G gives rise to a factor group G/H, and every factor group G/H gives rise to a homomorphism mapping G into G/H. Homomorphisms and factor groups are closely related. We give an example indicating how useful this relationship can be.

14.12 Example Classify the group $(\mathbb{Z}_4 \times \mathbb{Z}_2)/(\{0\} \times \mathbb{Z}_2)$ according to the fundamental theorem of finitely generated abelian groups (Theorem 11.12).

Solution The projection map $\pi_1 : \mathbb{Z}_4 \times \mathbb{Z}_2 \to \mathbb{Z}_4$ given by $\pi_1(x, y) = x$ is a homomorphism of $\mathbb{Z}_4 \times \mathbb{Z}_2$ onto $\mathbb{Z}_4$ with kernel $\{0\} \times \mathbb{Z}_2$. By Theorem 14.11, we know that the given factor group is isomorphic to $\mathbb{Z}_4$. ▲

Normal Subgroups and Inner Automorphisms

We derive some alternative characterizations of normal subgroups, which often provide us with an easier way to check normality than finding both the left and the right coset decompositions.

Suppose that H is a subgroup of G such that $ghg^{-1} \in H$ for all $g \in G$ and all $h \in H$. Then $gHg^{-1} = \{ghg^{-1} \mid h \in H\} \subseteq H$ for all $g \in G$. We claim that actually $gHg^{-1} = H$. We must show that $H \subseteq gHg^{-1}$ for all $g \in G$. Let $h \in H$. Replacing g by g^{-1} in the relation $ghg^{-1} \in H$, we obtain $g^{-1}h(g^{-1})^{-1} = g^{-1}hg = h_1$ where $h_1 \in H$. Consequently, $h = gh_1g^{-1} \in gHg^{-1}$, and we are done.

Suppose that $gH = Hg$ for all $g \in G$. Then $gh = h_1g$, so $ghg^{-1} \in H$ for all $g \in G$ and all $h \in H$. By the preceding paragraph, this means that $gHg^{-1} = H$ for all $g \in G$. Conversely, if $gHg^{-1} = H$ for all $g \in G$, then $ghg^{-1} = h_1$ so $gh = h_1g \in Hg$, and $gH \subseteq Hg$. But also, $g^{-1}Hg = H$ giving $g^{-1}hg = h_2$, so that $hg = gh_2$ and $Hg \subseteq gH$.

We summarize our work as a theorem.

14.13 Theorem The following are three equivalent conditions for a subgroup H of a group G to be a *normal* subgroup of G.

1. $ghg^{-1} \in H$ for all $g \in G$ and $h \in H$.
2. $gHg^{-1} = H$ for all $g \in G$.
3. $gH = Hg$ for all $g \in G$.

Condition (2) of Theorem 14.13 is often taken as the definition of a normal subgroup H of a group G.

14.14 Example Every subgroup H of an abelian group G is normal. We need only note that $gh = hg$ for all $h \in H$ and all $g \in G$, so, of course, $ghg^{-1} = h \in H$ for all $g \in G$ and all $h \in H$. ▲

Exercise 29 of Section 13 shows that the map $i_g : G \to G$ defined by $i_g(x) = gxg^{-1}$ is a homomorphism of G into itself. We see that $gag^{-1} = gbg^{-1}$ if and only if $a = b$, so i_g is one to one. Since $g(g^{-1}yg)g^{-1} = y$, we see that i_g is onto G, so it is an isomorphism of G with itself.

14.15 Definition An isomorphism $\phi : G \to G$ of a group G with itself is an **automorphism** of G. The automorphism $i_g : G \to G$, where $i_g(x) = gxg^{-1}$ for all $x \in G$, is the **inner automorphism of G by g**. Performing i_g on x is called **conjugation of x by g**. ■

The equivalence of conditions (1) and (2) in Theorem 14.13 shows that $gH = Hg$ for all $g \in G$ if and only if $i_g[H] = H$ for all $g \in G$, that is, if and only if H is **invariant** under all inner automorphisms of G. It is important to realize that $i_g[H] = H$ is an equation in *sets;* we need not have $i_g(h) = h$ for all $h \in H$. That is i_g may perform a nontrivial *permutation* of the set H. We see that the normal subgroups of a group G are precisely those that are invariant under all inner automorphisms. A subgroup K of G is a **conjugate subgroup** of H if $K = i_g[H]$ for some $g \in G$.

EXERCISES 14

Computations

In Exercises 1 through 8, find the order of the given factor group.

1. $\mathbb{Z}_6/\langle 3\rangle$

2. $(\mathbb{Z}_4 \times \mathbb{Z}_{12})/(\langle 2\rangle \times \langle 2\rangle)$

3. $(\mathbb{Z}_4 \times \mathbb{Z}_2)/\langle (2, 1)\rangle$

4. $(\mathbb{Z}_3 \times \mathbb{Z}_5)/(\{0\} \times \mathbb{Z}_5)$

5. $(\mathbb{Z}_2 \times \mathbb{Z}_4)/\langle (1, 1)\rangle$

6. $(\mathbb{Z}_{12} \times \mathbb{Z}_{18})/\langle (4, 3)\rangle$

7. $(\mathbb{Z}_2 \times S_3)/\langle (1, \rho_1)\rangle$

8. $(\mathbb{Z}_{11} \times \mathbb{Z}_{15})/\langle (1, 1)\rangle$

In Exercises 9 through 15, give the order of the element in the factor group.

9. $5 + \langle 4\rangle$ in $\mathbb{Z}_{12}/\langle 4\rangle$

10. $26 + \langle 12\rangle$ in $\mathbb{Z}_{60}/\langle 12\rangle$

11. $(2, 1) + \langle (1, 1)\rangle$ in $(\mathbb{Z}_3 \times \mathbb{Z}_6)/\langle (1, 1)\rangle$

12. $(3, 1) + \langle (1, 1)\rangle$ in $(\mathbb{Z}_4 \times \mathbb{Z}_4)/\langle (1, 1)\rangle$

13. $(3, 1) + \langle (0, 2)\rangle$ in $(\mathbb{Z}_4 \times \mathbb{Z}_8)/\langle (0, 2)\rangle$

14. $(3, 3) + \langle (1, 2)\rangle$ in $(\mathbb{Z}_4 \times \mathbb{Z}_8)/\langle (1, 2)\rangle$

15. $(2, 0) + \langle (4, 4)\rangle$ in $(\mathbb{Z}_6 \times \mathbb{Z}_8)/\langle (4, 4)\rangle$

16. Compute $i_{\rho_1}[H]$ for the subgroup $H = \{\rho_0, \mu_1\}$ of the group S_3 of Example 8.7.

Concepts

In Exercises 17 through 19, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

17. A *normal subgroup* H of G is one satisfying $hG = Gh$ for all $h \in H$.

18. A *normal subgroup* H of G is one satisfying $g^{-1}hg \in H$ for all $h \in H$ and all $g \in G$.

19. An *automorphism* of a group G is a homomorphism mapping G into G.

20. What is the importance of a *normal* subgroup of a group G?

Students often write nonsense when first proving theorems about factor groups. The next two exercises are designed to call attention to one basic type of error.

21. A student is asked to show that if H is a normal subgroup of an abelian group G, then G/H is abelian. The student's proof starts as follows:

We must show that G/H is abelian. Let a and b be two elements of G/H.

a. Why does the instructor reading this proof expect to find nonsense from here on in the student's paper?

b. What should the student have written?

c. Complete the proof.

22. A **torsion group** is a group all of whose elements have finite order. A group is **torsion free** if the identity is the only element of finite order. A student is asked to prove that if G is a torsion group, then so is G/H for every normal subgroup H of G. The student writes

We must show that each element of G/H is of finite order. Let $x \in G/H$.

Answer the same questions as in Exercise 21.

23. Mark each of the following true or false.

_____ **a.** It makes sense to speak of the factor group G/N if and only if N is a normal subgroup of the group G.

_____ **b.** Every subgroup of an abelian group G is a normal subgroup of G.

_____ **c.** An inner automorphism of an abelian group must be just the identity map.

_____ **d.** Every factor group of a finite group is again of finite order.

_____ **e.** Every factor group of a torsion group is a torsion group. (See Exercise 22.)

_____ **f.** Every factor group of a torsion-free group is torsion free. (See Exercise 22.)

_____ **g.** Every factor group of an abelian group is abelian.

_____ **h.** Every factor group of a nonabelian group is nonabelian.

_____ **i.** $\mathbb{Z}/n\mathbb{Z}$ is cyclic of order n.

_____ **j.** $\mathbb{R}/n\mathbb{R}$ is cyclic of order n, where $n\mathbb{R} = \{nr \mid r \in \mathbb{R}\}$ and $\mathbb{R}$ is under addition.

Theory

24. Show that A_n is a normal subgroup of S_n and compute S_n/A_n; that is, find a known group to which S_n/A_n is isomorphic.

25. Complete the proof of Theorem 14.4 by showing that if H is a subgroup of a group G and if left coset multiplication $(aH)(bH) = (ab)H$ is well defined, then $Ha \subseteq aH$.

26. Prove that the torsion subgroup T of an abelian group G is a normal subgroup of G, and that G/T is torsion free. (See Exercise 22.)

27. A subgroup H is **conjugate to a subgroup** K of a group G if there exists an inner automorphism i_g of G such that $i_g[H] = K$. Show that conjugacy is an equivalence relation on the collection of subgroups of G.

28. Characterize the normal subgroups of a group G in terms of the cells where they appear in the partition given by the conjugacy relation in the preceding exercise.

29. Referring to Exercise 27, find all subgroups of S_3 (Example 8.7) that are conjugate to $\{\rho_0, \mu_2\}$.

30. Let H be a normal subgroup of a group G, and let $m = (G : H)$. Show that $a^m \in H$ for every $a \in G$.

31. Show that an intersection of normal subgroups of a group G is again a normal subgroup of G.

32. Given any subset S of a group G, show that it makes sense to speak of the smallest normal subgroup that contains S. [*Hint:* Use Exercise 31.]

33. Let G be a group. An element of G that can be expressed in the form $aba^{-1}b^{-1}$ for some $a, b \in G$ is a **commutator** in G. The preceding exercise shows that there is a smallest normal subgroup C of a group G containing all commutators in G; the subgroup C is the **commutator subgroup** of G. Show that G/C is an abelian group.

34. Show that if a finite group G has exactly one subgroup H of a given order, then H is a normal subgroup of G.

35. Show that if H and N are subgroups of a group G, and N is normal in G, then $H \cap N$ is normal in H. Show by an example that $H \cap N$ need not be normal in G.

36. Let G be a group containing at least one subgroup of a fixed finite order s. Show that the intersection of all subgroups of G of order s is a normal subgroup of G. [*Hint:* Use the fact that if H has order s, then so does $x^{-1}Hx$ for all $x \in G$.]

37. a. Show that all automorphisms of a group G form a group under function composition.

b. Show that the inner automorphisms of a group G form a normal subgroup of the group of all automorphisms of G under function composition. [*Warning:* Be sure to show that the inner automorphisms do form a subgroup.]

38. Show that the set of all $g \in G$ such that $i_g : G \to G$ is the identity inner automorphism i_e is a normal subgroup of a group G.

39. Let G and G' be groups, and let H and H' be normal subgroups of G and G', respectively. Let ϕ be a homomorphism of G into G'. Show that ϕ induces a natural homomorphism $\phi_* : (G/H) \to (G'/H')$ if $\phi[H] \subseteq H'$. (This fact is used constantly in algebraic topology.)

40. Use the properties $\det(AB) = \det(A) \cdot \det(B)$ and $\det(I_n) = 1$ for $n \times n$ matrices to show the following:

a. The $n \times n$ matrices with determinant 1 form a normal subgroup of $GL(n, \mathbb{R})$.

b. The $n \times n$ matrices with determinant ± 1 form a normal subgroup of $GL(n, \mathbb{R})$.

41. Let G be a group, and let $\mathscr{P}(G)$ be the set of all *subsets* of G. For any $A, B \in \mathscr{P}(G)$, let us define the product subset $AB = \{ab \mid a \in A, b \in B\}$.

a. Show that this multiplication of subsets is associative and has an identity element, but that $\mathscr{P}(G)$ is not a group under this operation.

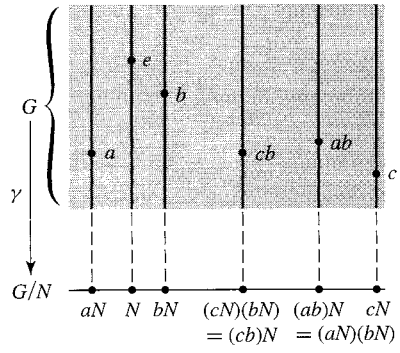
b. Show that if N is a normal subgroup of G, then the set of cosets of N is closed under the above operation on $\mathscr{P}(G)$, and that this operation agrees with the multiplication given by the formula in Corollary 14.5.

c. Show (without using Corollary 14.5) that the cosets of N in G form a group under the above operation. Is its identity element the same as the identity element of $\mathscr{P}(G)$?

SECTION 15 FACTOR-GROUP COMPUTATIONS AND SIMPLE GROUPS

Factor groups can be a tough topic for students to grasp. There is nothing like a bit of computation to strengthen understanding in mathematics. We start by attempting to improve our intuition concerning factor groups. Since we will be dealing with normal subgroups throughout this section, we often denote a subgroup of a group G by N rather than by H.

Let N be a normal subgroup of G. In the factor group G/N, the subgroup N acts as identity element. We may regard N as being *collapsed* to a single element, either to 0 in additive notation or to e in multiplicative notation. This collapsing of N together with the algebraic structure of G require that other subsets of G, namely, the cosets of N, also collapse into a single element in the factor group. A visualization of this collapsing is provided by Fig. 15.1. Recall from Theorem 14.9 that $\gamma : G \to G/N$ defined by $\gamma(a) = aN$ for $a \in G$ is a homomorphism of G onto G/N. Figure 15.1 is very similar to Fig. 13.14, but in Fig. 15.1 the image group under the homomorphism is actually formed from G. We can view the "line" G/N at the bottom of the figure as obtained by collapsing to a point each coset of N in another copy of G. Each point of G/N thus corresponds to a whole vertical line segment in the shaded portion, representing a coset of N in G. It is crucial to remember that multiplication of cosets in G/N can be computed by multiplying in G, using any representative elements of the cosets as shown in the figure.



15.1 Figure

Additively, two elements of G will collapse into the same element of G/N if they differ by an element of N. Multiplicatively, a and b collapse together if ab^{-1} is in N. The degree of collapsing can vary from nonexistent to catastrophic. We illustrate the two extreme cases by examples.

15.2 Example The trivial subgroup $N = \{0\}$ of $\mathbb{Z}$ is, of course, a normal subgroup. Compute $\mathbb{Z}/\{0\}$.

Solution Since $N = \{0\}$ has only one element, every coset of N has only one element. That is, the cosets are of the form $\{m\}$ for $m \in \mathbb{Z}$. There is no collapsing at all, and consequently, $\mathbb{Z}/\{0\} \simeq \mathbb{Z}$. Each $m \in \mathbb{Z}$ is simply renamed $\{m\}$ in $\mathbb{Z}/\{0\}$. ▲

15.3 Example Let n be a positive integer. The set $n\mathbb{R} = \{nr \mid r \in \mathbb{R}\}$ is a subgroup of $\mathbb{R}$ under addition, and it is normal since $\mathbb{R}$ is abelian. Compute $\mathbb{R}/n\mathbb{R}$.

Solution A bit of thought shows that actually $n\mathbb{R} = \mathbb{R}$, because each $x \in \mathbb{R}$ is of the form $n(x/n)$ and $x/n \in \mathbb{R}$. Thus $\mathbb{R}/n\mathbb{R}$ has only one element, the subgroup $n\mathbb{R}$. The factor group is a trivial group consisting only of the identity element. ▲

As illustrated in Examples 15.2 and 15.3 for any group G, we have $G/\{e\} \simeq G$ and $G/G \simeq \{e\}$, where $\{e\}$ is the trivial group consisting only of the identity element e. These two extremes of factor groups are of little importance. We would like knowledge of a factor group G/N to give some information about the structure of G. If $N = \{e\}$, the factor group has the same structure as G and we might as well have tried to study G directly. If $N = G$, the factor group has no significant structure to supply information about G. If G is a finite group and $N \neq \{e\}$ is a normal subgroup of G, then G/N is a smaller group than G, and consequently may have a more simple structure than G. The multiplication of cosets in G/N reflects the multiplication in G, since products of cosets can be computed by multiplying in G representative elements of the cosets.

We give two examples showing that even when G/N has order 2, we may be able to deduce some useful results. If G is a finite group and G/N has just two elements, then we must have $|G| = 2|N|$. Note that every subgroup H containing just half the elements of a finite group G must be a normal subgroup, since for each element a in G but not in H, both the left coset aH and the right coset Ha must consist of all elements in G that are not in H. Thus the left and right cosets of H coincide and H is a normal subgroup of G.

15.4 Example Because $|S_n| = 2|A_n|$, we see that A_n is a normal subgroup of S_n, and S_n/A_n has order 2. Let σ be an odd permutation in S_n, so that $S_n/A_n = \{A_n, \sigma A_n\}$. Renaming the element A_n "even" and the element σA_n "odd," the multiplication in S_n/A_n shown in Table 15.5 becomes

15.5 Table

	A_n	σA_n
A_n	A_n	σA_n
σA_n	σA_n	A_n

$$\begin{array}{ll} (\text{even})(\text{even}) = \text{even} & (\text{odd})(\text{even}) = \text{odd} \\ (\text{even})(\text{odd}) = \text{odd} & (\text{odd})(\text{odd}) = \text{even}. \end{array}$$

Thus the factor group reflects these multiplicative properties for all the permutations in S_n. ▲

Example 15.4 illustrates that while knowing the product of two cosets in G/N does not tell us what the product of two elements of G is, it may tell us that the product in G of two *types* of elements is itself of a certain type.

15.6 Example **(Falsity of the Converse of the Theorem of Lagrange)** The theorem of Lagrange states if H is a subgroup of a finite group G, then the order of H divides the order of G. We show that it is false that if d divides the order of G, then there must exist a subgroup H of G having order d. Namely, we show that A_4, which has order 12, contains no subgroup of order 6.

Suppose that H were a subgroup of A_4 having order 6. As observed before in Example 15.4, it would follow that H would be a normal subgroup of A_4. Then A_4/H would have only two elements, H and σH for some $\sigma \in A_4$ not in H. Since in a group of order 2, the square of each element is the identity, we would have $HH = H$ and $(\sigma H)(\sigma H) = H$. Now computation in a factor group can be achieved by computing with representatives in the original group. Thus, computing in A_4, we find that for each $\alpha \in H$ we must have $\alpha^2 \in H$ and for each $\beta \in \sigma H$ we must have $\beta^2 \in H$. That is, the square of every element in A_4 must be in H. But in A_4, we have

$$(1, 2, 3) = (1, 3, 2)^2 \quad \text{and} \quad (1, 3, 2) = (1, 2, 3)^2$$

so (1, 2, 3) and (1, 3, 2) are in H. A similar computation shows that (1, 2, 4), (1, 4, 2), (1, 3, 4), (1, 4, 3), (2, 3, 4), and (2, 4, 3) are all in H. This shows that there must be at least 8 elements in H, contradicting the fact that H was supposed to have order 6. ▲

We now turn to several examples that *compute* factor groups. If the group we start with is finitely generated and abelian, then its factor group will be also. *Computing* such a factor group means classifying it according to the fundamental theorem (Theorem 11.12).

15.7 Example Let us compute the factor group $(\mathbb{Z}_4 \times \mathbb{Z}_6)/\langle(0, 1)\rangle$. Here $\langle(0, 1)\rangle$ is the cyclic subgroup H of $\mathbb{Z}_4 \times \mathbb{Z}_6$ generated by (0, 1). Thus

$$H = \{(0, 0), (0, 1), (0, 2), (0, 3), (0, 4), (0, 5)\}.$$

Since $\mathbb{Z}_4 \times \mathbb{Z}_6$ has 24 elements and H has 6 elements, all cosets of H must have 6 elements, and $(\mathbb{Z}_4 \times \mathbb{Z}_6)/H$ must have order 4. Since $\mathbb{Z}_4 \times \mathbb{Z}_6$ is abelian, so is $(\mathbb{Z}_4 \times \mathbb{Z}_6)/H$ (remember, we compute in a factor group by means of representatives from the original group). In additive notation, the cosets are

$$H = (0, 0) + H, \qquad (1, 0) + H, \qquad (2, 0) + H, \qquad (3, 0) + H.$$

Since we can compute by choosing the representatives (0, 0), (1, 0), (2, 0), and (3, 0), it is clear that $(\mathbb{Z}_4 \times \mathbb{Z}_6)/H$ is isomorphic to $\mathbb{Z}_4$. Note that this is what we would expect, since in a factor group modulo H, everything in H becomes the identity element; that is, we are essentially setting everything in H equal to zero. Thus the whole second factor $\mathbb{Z}_6$ of $\mathbb{Z}_4 \times \mathbb{Z}_6$ is collapsed, leaving just the first factor $\mathbb{Z}_4$. ▲

Example 15.7 is a special case of a general theorem that we now state and prove. We should acquire an intuitive feeling for this theorem in terms of *collapsing one of the factors to the identity element.*

15.8 Theorem Let $G = H \times K$ be the direct product of groups H and K. Then $\overline{H} = \{(h, e) \mid h \in H\}$ is a normal subgroup of G. Also $G/\overline{H}$ is isomorphic to K in a natural way. Similarly, $G/\overline{K} \simeq H$ in a natural way.

Proof Consider the homomorphism $\pi_2 : H \times K \to K$, where $\pi_2(h, k) = k$. (See Example 13.8). Because $\text{Ker}(\pi_2) = \overline{H}$, we see that $\overline{H}$ is a normal subgroup of $H \times K$. Because π_2 is onto K, Theorem 14.11 tells us that $(H \times K)/\overline{H} \simeq K$. ◆

We continue with additional computations of abelian factor groups. To illustrate how easy it is to compute in a factor group if we can compute in the whole group, we prove the following theorem.

15.9 Theorem A factor group of a cyclic group is cyclic.

Proof Let G be cyclic with generator a, and let N be a normal subgroup of G. We claim the coset aN generates G/N. We must compute all powers of aN. But this amounts to computing, in G, all powers of the representative a and all these powers give all elements in G. Hence the powers of aN certainly give all cosets of N and G/N is cyclic. ◆

15.10 Example Let us compute the factor group $(\mathbb{Z}_4 \times \mathbb{Z}_6)/\langle(0, 2)\rangle$. Now $(0, 2)$ generates the subgroup

$$H = \{(0, 0), (0, 2), (0, 4)\}$$

of $\mathbb{Z}_4 \times \mathbb{Z}_6$ of order 3. Here the first factor $\mathbb{Z}_4$ of $\mathbb{Z}_4 \times \mathbb{Z}_6$ is left alone. The $\mathbb{Z}_6$ factor, on the other hand, is essentially collapsed by a subgroup of order 3, giving a factor group in the second factor of order 2 that must be isomorphic to $\mathbb{Z}_2$. Thus $(\mathbb{Z}_4 \times \mathbb{Z}_6)/\langle(0, 2)\rangle$ is isomorphic to $\mathbb{Z}_4 \times \mathbb{Z}_2$. ▲

15.11 Example Let us compute the factor group $(\mathbb{Z}_4 \times \mathbb{Z}_6)/\langle(2, 3)\rangle$. *Be careful*! There is a great temptation to say that we are setting the 2 of $\mathbb{Z}_4$ and the 3 of $\mathbb{Z}_6$ both equal to zero, so that $\mathbb{Z}_4$ is collapsed to a factor group isomorphic to $\mathbb{Z}_2$ and $\mathbb{Z}_6$ to one isomorphic to $\mathbb{Z}_3$, giving a total factor group isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_3$. *This is wrong*! Note that

$$H = \langle(2, 3)\rangle = \{(0, 0), (2, 3)\}$$

is of order 2, so $(\mathbb{Z}_4 \times \mathbb{Z}_6)/\langle(2, 3)\rangle$ has order 12, not 6. Setting $(2, 3)$ equal to zero does not make $(2, 0)$ and $(0, 3)$ equal to zero individually, so the factors do not collapse separately.

The possible abelian groups of order 12 are $\mathbb{Z}_4 \times \mathbb{Z}_3$ and $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3$, and we must decide to which one our factor group is isomorphic. These two groups are most easily distinguished in that $\mathbb{Z}_4 \times \mathbb{Z}_3$ has an element of order 4, and $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3$ does not. We claim that the coset $(1, 0) + H$ is of order 4 in the factor group $(\mathbb{Z}_4 \times \mathbb{Z}_6)/H$. To find the smallest power of a coset giving the identity in a factor group modulo H, we must, by choosing representatives, find the smallest power of a representative that is in the subgroup H. Now,

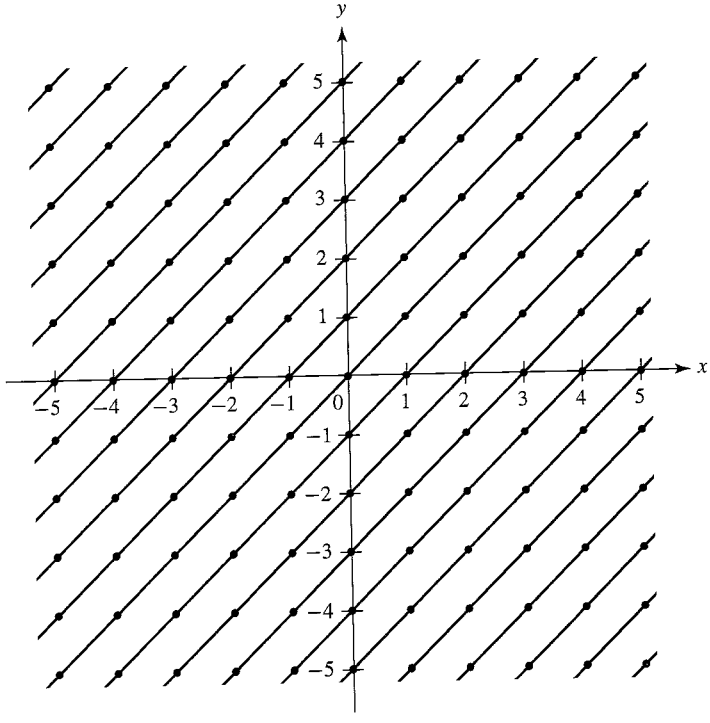
$$4(1, 0) = (1, 0) + (1, 0) + (1, 0) + (1, 0) = (0, 0)$$

is the first time that $(1, 0)$ added to itself gives an element of H. Thus $(\mathbb{Z}_4 \times \mathbb{Z}_6)/\langle(2, 3)\rangle$ has an element of order 4 and is isomorphic to $\mathbb{Z}_4 \times \mathbb{Z}_3$ or $\mathbb{Z}_{12}$. ▲

15.12 Example Let us compute (that is, classify as in Theorem 11.12 the group $(\mathbb{Z} \times \mathbb{Z})/\langle(1, 1)\rangle$. We may visualize $\mathbb{Z} \times \mathbb{Z}$ as the points in the plane with both coordinates integers, as indicated by the dots in Fig. 15.13. The subgroup $\langle(1, 1)\rangle$ consists of those points that lie on the 45° line through the origin, indicated in the figure. The coset $(1, 0) + \langle(1, 1)\rangle$ consists of those dots on the 45° line through the point $(1, 0)$, also shown in the figure. Continuing, we see that each coset consists of those dots lying on one of the 45° lines in the figure. We may choose the representatives

$$\cdots, (-3, 0), (-2, 0), (-1, 0), (0, 0), (1, 0), (2, 0), (3, 0), \cdots$$

of these cosets to compute in the factor group. Since these representatives correspond precisely to the points of $\mathbb{Z}$ on the x-axis, we see that the factor group $(\mathbb{Z} \times \mathbb{Z})/\langle(1, 1)\rangle$ is isomorphic to $\mathbb{Z}$. ▲



15.13 Figure

Simple Groups

As we mentioned in the preceding section, one feature of a factor group is that it gives crude information about the structure of the whole group. Of course, sometimes there may be no nontrivial proper normal subgroups. For example, Theorem 10.10 shows that a group of prime order can have no nontrivial proper subgroups of any sort.

15.14 Definition A group is **simple** if it is nontrivial and has no proper nontrivial normal subgroups. ■

15.15 Theorem The alternating group A_n is simple for $n \geq 5$.

Proof See Exercise 39. ◆

There are many simple groups other than those given above. For example, A_5 is of order 60 and A_6 is of order 360, and there is a simple group of nonprime order, namely 168, between these orders.

The complete determination and classification of all finite simple groups were recently completed. Hundreds of mathematicians worked on this task from 1950 to 1980. It can be shown that a finite group has a sort of factorization into simple groups, where the factors are unique up to order. The situation is similar to the factorization of positive integers into primes. The new knowledge of all finite simple groups can now be used to solve some problems of finite group theory.

We have seen in this text that a finite simple abelian group is isomorphic to $\mathbb{Z}_p$ for some prime p. In 1963, Thompson and Feit [21] published their proof of a longstanding conjecture of Burnside, showing that every finite nonabelian simple group is of even order. Further great strides toward the complete classification were made by Aschbacher in the 1970s. Early in 1980, Griess announced that he had constructed a predicted "monster" simple group of order

$$808,017,424,794,512,875,886,459,904,961,710,757,005,754,368,$$
$$000,000,000.$$

Aschbacher added the final details of the classification in August 1980. The research papers contributing to the entire classification fill roughly 5000 journal pages.

We turn to the characterization of those normal subgroups N of a group G for which G/N is a simple group. First we state an addendum to Theorem 13.12 on properties of a group homomorphism. The proof is left to Exercises 35 and 36.

15.16 Theorem Let $\phi : G \to G'$ be a group homomorphism. If N is a normal subgroup of G, then $\phi[N]$ is a normal subgroup of $\phi[G]$. Also, if N' is a normal subgroup of $\phi[G]$, then $\phi^{-1}[N']$ is a normal subgroup of G.

Theorem 15.16 should be viewed as saying that a homomorphism $\phi : G \to G'$ preserves normal subgroups between G and $\phi[G]$. It is important to note that $\phi[N]$ may not be normal in G', even though N is normal in G. For example, $\phi : \mathbb{Z}_2 \to S_3$, where $\phi(0) = \rho_0$ and $\phi(1) = \mu_1$ is a homomorphism, and $\mathbb{Z}_2$ is a normal subgroup of itself, but $\{\rho_0, \mu_1\}$ is not a normal subgroup of S_3.

We can now characterize when G/N is a simple group.

15.17 Definition A **maximal normal subgroup of a group** G is a normal subgroup M not equal to G such that there is no proper normal subgroup N of G properly containing M. ■

15.18 Theorem M is a maximal normal subgroup of G if and only if G/M is simple.

Proof Let M be a maximal normal subgroup of G. Consider the canonical homomorphism $\gamma : G \to G/M$ given by Theorem 14.9. Now γ^{-1} of any nontrivial proper normal subgroup of G/M is a proper normal subgroup of G properly containing M. But M is maximal, so this can not happen. Thus G/M is simple.

Conversely, Theorem 15.16 shows that if N is a normal subgroup of G properly containing M, then $\gamma[N]$ is normal in G/M. If also $N \neq G$, then

$$\gamma[N] \neq G/M \qquad \text{and} \qquad \gamma[N] \neq \{M\}.$$

Thus, if G/M is simple so that no such $\gamma[N]$ can exist, no such N can exist, and M is maximal. ◆

The Center and Commutator Subgroups

Every nonabelian group G has two important normal subgroups, the *center* $Z(G)$ of G and the *commutator subgroup* C of G. (The letter Z comes from the German word *zentrum,* meaning center.) The center $Z(G)$ is defined by

$$Z(G) = \{z \in G \mid zg = gz \text{ for all } g \in G\}.$$

Exercise 52 of Section 5 shows that $Z(G)$ is an abelian subgroup of G. Since for each $g \in G$ and $z \in Z(G)$ we have $gzg^{-1} = zgg^{-1} = ze = z$, we see at once that $Z(G)$ is a normal subgroup of G. If G is abelian, then $Z(G) = G$; in this case, the center is not useful.

15.19 Example The center of a group G always contains the identity element e. It may be that $Z(G) = \{e\}$, in which case we say that **the center of G is trivial.** For example, examination of Table 8.8 for the group S_3 shows us that $Z(S_3) = \{\rho_0\}$, so the center of S_3 is trivial. (This is a special case of Exercise 38, which shows that the center of every nonabelian group of order pq for primes p and q is trivial.) Consequently, the center of $S_3 \times \mathbb{Z}_5$ must be $\{\rho_0\} \times \mathbb{Z}_5$, which is isomorphic to $\mathbb{Z}_5$. ▲

Turning to the commutator subgroup, recall that in forming a factor group of G modulo a normal subgroup N, we are essentially putting every element in G that is in N equal to e, for N forms our new identity in the factor group. This indicates another use for factor groups. Suppose, for example, that we are studying the structure of a nonabelian group G. Since Theorem 11.12 gives complete information about the structure of all sufficiently small abelian groups, it might be of interest to try to form an abelian group as much like G as possible, an *abelianized version* of G, by starting with G and then requiring that $ab = ba$ for all a and b in our new group structure. To require that $ab = ba$ is to say that $aba^{-1}b^{-1} = e$ in our new group. An element $aba^{-1}b^{-1}$ in a group is a **commutator of the group.** Thus we wish to attempt to form an abelianized version of G by replacing every commutator of G by e. By the first observation of this paragraph, we should then attempt to form the factor group of G modulo the smallest normal subgroup we can find that contains all commutators of G.

15.20 Theorem Let G be a group. The set of all commutators $aba^{-1}b^{-1}$ for $a, b \in G$ generates a subgroup C (the **commutator subgroup**) of G. This subgroup C is a normal subgroup of G. Furthermore, if N is a normal subgroup of G, then G/N is abelian if and only if $C \leq N$.

Proof The commutators certainly generate a subgroup C; we must show that it is normal in G. Note that the inverse $(aba^{-1}b^{-1})^{-1}$ of a commutator is again a commutator, namely, $bab^{-1}a^{-1}$. Also $e = eee^{-1}e^{-1}$ is a commutator. Theorem 7.6 then shows that C consists precisely of all finite products of commutators. For $x \in C$, we must show that $g^{-1}xg \in C$ for all $g \in G$, or that if x is a product of commutators, so is $g^{-1}xg$ for all $g \in G$. By inserting $e = gg^{-1}$ between each product of commutators occurring in x, we see that it is sufficient to show for each commutator $cdc^{-1}d^{-1}$ that $g^{-1}(cdc^{-1}d^{-1})g$ is in C. But

$$\begin{aligned} g^{-1}(cdc^{-1}d^{-1})g &= (g^{-1}cdc^{-1})(e)(d^{-1}g) \\ &= (g^{-1}cdc^{-1})(gd^{-1}dg^{-1})(d^{-1}g) \\ &= [(g^{-1}c)d(g^{-1}c)^{-1}d^{-1}][dg^{-1}d^{-1}g], \end{aligned}$$

which is in C. Thus C is normal in G.

The rest of the theorem is obvious if we have acquired the proper feeling for factor groups. One doesn't visualize in this way, but writing out that G/C is abelian follows from

$$\begin{aligned} (aC)(bC) &= abC = ab(b^{-1}a^{-1}ba)C \\ &= (abb^{-1}a^{-1})baC = baC = (bC)(aC). \end{aligned}$$

Furthermore, if N is a normal subgroup of G and G/N is abelian, then $(a^{-1}N)(b^{-1}N) = (b^{-1}N)(a^{-1}N)$; that is, $aba^{-1}b^{-1}N = N$, so $aba^{-1}b^{-1} \in N$, and $C \leq N$. Finally, if $C \leq N$, then

$$\begin{aligned} (aN)(bN) &= abN = ab(b^{-1}a^{-1}ba)N \\ &= (abb^{-1}a^{-1})baN = baN = (bN)(aN). \end{aligned}$$

◆

15.21 Example For the group S_3 in Table 8.8, we find that one commutator is $\rho_1\mu_1\rho_1^{-1}\mu_1^{-1} = \rho_1\mu_1\rho_2\mu_1 = \mu_3\mu_2 = \rho_2$. We similarly find that $\rho_2\mu_1\rho_2^{-1}\mu_1^{-1} = \rho_2\mu_1\rho_1\mu_1 = \mu_2\mu_3 = \rho_1$. Thus the commutator subgroup C of S_3 contains A_3. Since A_3 is a normal subgroup of S_3 and S_3/A_3 is abelian, Theorem 15.20 shows that $C = A_3$. ▲

EXERCISES 15

Computations

In Exercises 1 through 12, classify the given group according to the fundamental theorem of finitely generated abelian groups.

1. $(\mathbb{Z}_2 \times \mathbb{Z}_4)/\langle(0, 1)\rangle$

2. $(\mathbb{Z}_2 \times \mathbb{Z}_4)/\langle(0, 2)\rangle$

3. $(\mathbb{Z}_2 \times \mathbb{Z}_4)/\langle(1, 2)\rangle$

4. $(\mathbb{Z}_4 \times \mathbb{Z}_8)/\langle(1, 2)\rangle$

5. $(\mathbb{Z}_4 \times \mathbb{Z}_4 \times \mathbb{Z}_8)/\langle(1, 2, 4)\rangle$

6. $(\mathbb{Z} \times \mathbb{Z})/\langle(0, 1)\rangle$

7. $(\mathbb{Z} \times \mathbb{Z})/\langle(1, 2)\rangle$

8. $(\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z})/\langle(1, 1, 1)\rangle$

9. $(\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}_4)/\langle(3, 0, 0)\rangle$

10. $(\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}_8)/\langle(0, 4, 0)\rangle$

11. $(\mathbb{Z} \times \mathbb{Z})/\langle(2, 2)\rangle$

12. $(\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z})/\langle(3, 3, 3)\rangle$

13. Find both the center $Z(D_4)$ and the commutator subgroup C of the group D_4 of symmetries of the square in Table 8.12.

14. Find both the center and the commutator subgroup of $\mathbb{Z}_3 \times S_3$.

15. Find both the center and the commutator subgroup of $S_3 \times D_4$.

16. Describe all subgroups of order ≤ 4 of $\mathbb{Z}_4 \times \mathbb{Z}_4$, and in each case classify the factor group of $\mathbb{Z}_4 \times \mathbb{Z}_4$ modulo the subgroup by Theorem 11.12. That is, describe the subgroup and say that the factor group of $\mathbb{Z}_4 \times \mathbb{Z}_4$ modulo the subgroup is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_4$, or whatever the case may be. [*Hint:* $\mathbb{Z}_4 \times \mathbb{Z}_4$ has six different cyclic subgroups of order 4. Describe them by giving a generator, such as the subgroup $\langle (1, 0) \rangle$. There is one subgroup of order 4 that is isomorphic to the Klein 4-group. There are three subgroups of order 2.]

Concepts

In Exercises 17 and 18, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

17. The *center* of a group G contains all elements of G that commute with every element of G.

18. The *commutator subgroup* of a group G is $\{a^{-1}b^{-1}ab \mid a, b \in G\}$.

19. Mark each of the following true or false.

_____ **a.** Every factor group of a cyclic group is cyclic.
_____ **b.** A factor group of a noncyclic group is again noncyclic.
_____ **c.** $\mathbb{R}/\mathbb{Z}$ under addition has no element of order 2.
_____ **d.** $\mathbb{R}/\mathbb{Z}$ under addition has elements of order n for all $n \in \mathbb{Z}^+$.
_____ **e.** $\mathbb{R}/\mathbb{Z}$ under addition has an infinite number of elements of order 4.
_____ **f.** If the commutator subgroup C of a group G is $\{e\}$, then G is abelian.
_____ **g.** If G/H is abelian, then the commutator subgroup of C of G contains H.
_____ **h.** The commutator subgroup of a simple group G must be G itself.
_____ **i.** The commutator subgroup of a nonabelian simple group G must be G itself.
_____ **j.** All nontrivial finite simple groups have prime order.

In Exercises 20 through 23, let F be the additive group of all functions mapping $\mathbb{R}$ into $\mathbb{R}$, and let F^* be the multiplicative group of all elements of F that do not assume the value 0 at any point of $\mathbb{R}$.

20. Let K be the subgroup of F consisting of the constant functions. Find a subgroup of F to which F/K is isomorphic.

21. Let K^* be the subgroup of F^* consisting of the nonzero constant functions. Find a subgroup of F^* to which F^*/K^* is isomorphic.

22. Let K be the subgroup of continuous functions in F. Can you find an element of F/K having order 2? Why or why not?

23. Let K^* be the subgroup of F^* consisting of the continuous functions in F^*. Can you find an element of F^*/K^* having order 2? Why or why not?

In Exercises 24 through 26, let U be the multiplicative group $\{z \in \mathbb{C} \mid |z| = 1\}$.

24. Let $z_0 \in U$. Show that $z_0U = \{z_0 z \mid z \in U\}$ is a subgroup of U, and compute U/z_0U.

25. To what group we have mentioned in the text is $U/\langle -1 \rangle$ isomorphic?

26. Let $\zeta_n = \cos(2\pi/n) + i \sin(2\pi/n)$ where $n \in \mathbb{Z}^+$. To what group we have mentioned is $U/\langle \zeta_n \rangle$ isomorphic?

27. To what group mentioned in the text is the additive group $\mathbb{R}/\mathbb{Z}$ isomorphic?

28. Give an example of a group G having no elements of finite order > 1 but having a factor group G/H, all of whose elements are of finite order.

29. Let H and K be normal subgroups of a group G. Give an example showing that we may have $H \simeq K$ while G/H is not isomorphic to G/K.

30. Describe the center of every simple

a. abelian group

b. nonabelian group.

31. Describe the commutator subgroup of every simple

a. abelian group

b. nonabelian group.

Proof Synopsis

32. Give a one-sentence synopsis of the proof of Theorem 15.9.

33. Give at most a two-sentence synopsis of the proof of Theorem 15.18.

Theory

34. Show that if a finite group G contains a nontrivial subgroup of index 2 in G, then G is not simple.

35. Let $\phi : G \to G'$ be a group homomorphism, and let N be a normal subgroup of G. Show that $\phi[N]$ is normal subgroup of $\phi[G]$.

36. Let $\phi : G \to G'$ be a group homomorphism, and let N' be a normal subgroup of G'. Show that $\phi^{-1}[N']$ is a normal subgroup of G.

37. Show that if G is nonabelian, then the factor group $G/Z(G)$ is not cyclic. [*Hint:* Show the equivalent contrapositive, namely, that if $G/Z(G)$ is cyclic then G is abelian (and hence $Z(G) = G$).]

38. Using Exercise 37, show that a nonabelian group G of order pq where p and q are primes has a trivial center.

39. Prove that A_n is simple for $n \geq 5$, following the steps and hints given.

a. Show A_n contains every 3-cycle if $n \geq 3$.

b. Show A_n is generated by the 3-cycles for $n \geq 3$. [*Hint:* Note that $(a, b)(c, d) = (a, c, b)(a, c, d)$ and $(a, c)(a, b) = (a, b, c)$.]

c. Let r and s be fixed elements of $\{1, 2, \cdots, n\}$ for $n \geq 3$. Show that A_n is generated by the n "special" 3-cycles of the form (r, s, i) for $1 \leq i \leq n$ [*Hint:* Show every 3-cycle is the product of "special" 3-cycles by computing

$$(r, s, i)^2, \quad (r, s, j)(r, s, i)^2, \quad (r, s, j)^2(r, s, i),$$

and

$$(r, s, i)^2(r, s, k)(r, s, j)^2(r, s, i).$$

Observe that these products give all possible types of 3-cycles.]

d. Let N be a normal subgroup of A_n for $n \geq 3$. Show that if N contains a 3-cycle, then $N = A_n$. [*Hint:* Show that $(r, s, i) \in N$ implies that $(r, s, j) \in N$ for $j = 1, 2, \cdots, n$ by computing

$$((r, s)(i, j))(r, s, i)^2((r, s)(i, j))^{-1}.]$$

e. Let N be a nontrivial normal subgroup of A_n for $n \geq 5$. Show that one of the following cases must hold, and conclude in each case that $N = A_n$.

Case I N contains a 3-cycle.

Case II N contains a product of disjoint cycles, at least one of which has length greater than 3. [*Hint:* Suppose N contains the disjoint product $\sigma = \mu(a_1, a_2, \cdots, a_r)$. Show $\sigma^{-1}(a_1, a_2, a_3)\sigma(a_1, a_2, a_3)^{-1}$ is in N, and compute it.]

Case III N contains a disjoint product of the form $\sigma = \mu(a_4, a_5, a_6)(a_1, a_2, a_3)$. [*Hint:* Show $\sigma^{-1}(a_1, a_2, a_4)\sigma(a_1, a_2, a_4)^{-1}$ is in N, and compute it.]

Case IV N contains a disjoint product of the form $\sigma = \mu(a_1, a_2, a_3)$ where μ is a product of disjoint 2-cycles. [*Hint:* Show $\sigma^2 \in N$ and compute it.]

Case V N contains a disjoint product σ of the form $\sigma = \mu(a_3, a_4)(a_1, a_2)$, where μ is a product of an even number of disjoint 2-cycles. [*Hint:* Show that $\sigma^{-1}(a_1, a_2, a_3)\sigma(a_1, a_2, a_3)^{-1}$ is in N, and compute it to deduce that $\alpha = (a_2, a_4)(a_1, a_3)$ is in N. Using $n \geq 5$ for the first time, find $i \neq a_1, a_2, a_3, a_4$ in $\{1, 2, \cdots, n\}$. Let $\beta = (a_1, a_3, i)$. Show that $\beta^{-1}\alpha\beta\alpha \in N$, and compute it.]

40. Let N be a normal subgroup of G and let H be any subgroup of G. Let $HN = \{hn \mid h \in H, n \in N\}$. Show that HN is a subgroup of G, and is the smallest subgroup containing both N and H.

41. With reference to the preceding exercise, let M also be a normal subgroup of G. Show that NM is again a normal subgroup of G.

42. Show that if H and K are normal subgroups of a group G such that $H \cap K = \{e\}$, then $hk = kh$ for all $h \in H$ and $k \in K$. [*Hint:* Consider the commutator $hkh^{-1}k^{-1} = (hkh^{-1})k^{-1} = h(kh^{-1}k^{-1})$.]

SECTION 16 †GROUP ACTION ON A SET

We have seen examples of how groups may *act on things*, like the group of symmetries of a triangle or of a square, the group of rotations of a cube, the general linear group acting on $\mathbb{R}^n$, and so on. In this section, we give the general notion of group action on a set. The next section will give an application to counting.

The Notion of a Group Action

Definition 2.1 defines a binary operation $*$ on a set S to be a function mapping $S \times S$ into S. The function $*$ gives us a rule for "multiplying" an element s_1 in S and an element s_2 in S to yield an element $s_1 * s_2$ in S.

More generally, for any sets A, B, and C, we can view a map $* : A \times B \to C$ as defining a "multiplication," where any element a of A times any element b of B has as value some element c of C. Of course, we write $a * b = c$, or simply $ab = c$. In this section, we will be concerned with the case where X is a set, G is a group, and we have a map $* : G \times X \to X$. We shall write $*(g, x)$ as $g * x$ or gx.

16.1 Definition Let X be a set and G a group. An **action of G on X** is a map $* : G \times X \to X$ such that ■

1. $ex = x$ for all $x \in X$,
2. $(g_1g_2)(x) = g_1(g_2x)$ for all $x \in X$ and all $g_1, g_2 \in G$.

Under these conditions, X is a **G-set.**

† This section is a prerequisite only for Sections 17 and 36.

16.2 Example Let X be any set, and let H be a subgroup of the group S_X of all permutations of X. Then X is an H-set, where the action of $\sigma \in H$ on X is its action as an element of S_X, so that $\sigma x = \sigma(x)$ for all $x \in X$. Condition 2 is a consequence of the definition of permutation multiplication as function composition, and Condition 1 is immediate from the definition of the identity permutation as the identity function. Note that, in particular, $\{1, 2, 3, \cdots, n\}$ is an S_n set. ▲

Our next theorem will show that for every G-set X and each $g \in G$, the map $\sigma_g : X \to X$ defined by $\sigma_g(x) = gx$ is a permutation of X, and that there is a homomorphism $\phi : G \to S_X$ such that the action of G on X is essentially the Example 16.2 action of the image subgroup $H = \phi[G]$ of S_X on X. So actions of subgroups of S_X on X describe all possible group actions on X. When studying the set X, actions using subgroups of S_X suffice. However, sometimes a set X is used to study G via a group action of G on X. Thus we need the more general concept given by Definition 16.1.

16.3 Theorem Let X be a G-set. For each $g \in G$, the function $\sigma_g : X \to X$ defined by $\sigma_g(x) = gx$ for $x \in X$ is a permutation of X. Also, the map $\phi : G \to S_X$ defined by $\phi(g) = \sigma_g$ is a homomorphism with the property that $\phi(g)(x) = gx$.

Proof To show that σ_g is a permutation of X, we must show that it is a one-to-one map of X onto itself. Suppose that $\sigma_g(x_1) = \sigma_g(x_2)$ for $x_1, x_2 \in X$. Then $gx_1 = gx_2$. Consequently, $g^{-1}(gx_1) = g^{-1}(gx_2)$. Using Condition 2 in Definition 16.1, we see that $(g^{-1}g)x_1 = (g^{-1}g)x_2$, so $ex_1 = ex_2$. Condition 1 of the definition then yields $x_1 = x_2$, so σ_g is one to one. The two conditions of the definition show that for $x \in X$, we have $\sigma_g(g^{-1}x) = g(g^{-1})x = (gg^{-1})x = ex = x$, so σ_g maps X onto X. Thus σ_g is indeed a permutation.

To show that $\phi : G \to S_X$ defined by $\phi(g) = \sigma_g$ is a homomorphism, we must show that $\phi(g_1g_2) = \phi(g_1)\phi(g_2)$ for all $g_1, g_2 \in G$. We show the equality of these two permutations in S_X by showing they both carry an $x \in X$ into the same element. Using the two conditions in Definition 16.1 and the rule for function composition, we obtain

$$\begin{aligned}\phi(g_1g_2)(x) &= \sigma_{g_1g_2}(x) = (g_1g_2)x = g_1(g_2x) = g_1\sigma_{g_2}(x) = \sigma_{g_1}(\sigma_{g_2}(x)) \\ &= (\sigma_{g_1} \circ \sigma_{g_2})(x) = (\sigma_{g_1}\sigma_{g_2})(x) = (\phi(g_1)\phi(g_2))(x).\end{aligned}$$

Thus ϕ is a homomorphism. The stated property of ϕ follows at once since by our definitions, we have $\phi(g)(x) = \sigma_g(x) = gx$. ◆

It follows from the preceding theorem and Theorem 13.15 that if X is G-set, then the subset of G leaving every element of X fixed is a normal subgroup N of G, and we can regard X as a G/N-set where the action of a coset gN on X is given by $(gN)x = gx$ for each $x \in X$. If $N = \{e\}$, then the identity element of G is the only element that leaves every $x \in X$ fixed; we then say that G **acts faithfully** on X. A group G is **transitive** on a G-set X if for each $x_1, x_2 \in X$, there exists $g \in G$ such that $gx_1 = x_2$. Note that G is transitive on X if and only if the subgroup $\phi[G]$ of S_X is transitive on X, as defined in Exercise 49 of Section 8.

We continue with more examples of G-sets.

16.4 Example Every group G is itself a G-set, where the action on $g_2 \in G$ by $g_1 \in G$ is given by left multiplication. That is, $*(g_1, g_2) = g_1 g_2$. If H is a subgroup of G, we can also regard G as an H-set, where $*(h, g) = hg$. ▲

16.5 Example Let H be a subgroup of G. Then G is an H-set under conjugation where $*(h, g) = hgh^{-1}$ for $g \in G$ and $h \in H$. Condition 1 is obvious, and for Condition 2 note that

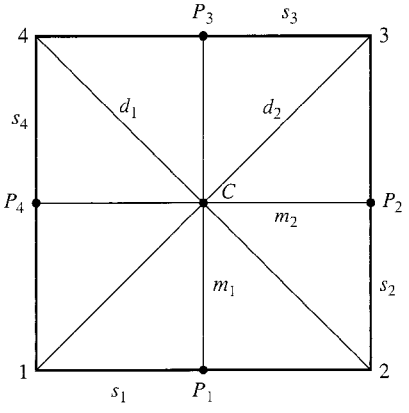
$$*(h_1h_2, g) = (h_1h_2)g(h_1h_2)^{-1} = h_1(h_2gh_2^{-1})h_1^{-1} = *(h_1, *(h_2, g)).$$

We always write this action of H on G by conjugation as hgh^{-1}. The abbreviation hg described before the definition would cause terrible confusion with the group operation of G. ▲

16.6 Example For students who have studied vector spaces with real (or complex) scalars, we mention that the axioms $(rs)\mathbf{v} = r(s\mathbf{v})$ and $1\mathbf{v} = \mathbf{v}$ for scalars r and s and a vector $\mathbf{v}$ show that the set of vectors is an R^*-set (or a C^*-set) for the multiplicative group of nonzero scalars. ▲

16.7 Example Let H be a subgroup of G, and let L_H be the set of all left cosets of H. Then L_H is a G-set, where the action of $g \in G$ on the left coset xH is given by $g(xH) = (gx)H$. Observe that this action is well defined: if $yH = xH$, then $y = xh$ for some $h \in H$, and $g(yH) = (gy)H = (gxh)H = (gx)(hH) = (gx)H = g(xH)$. A series of exercises shows that every G-set is isomorphic to one that may be formed using these left coset G-sets as building blocks. (See Exercises 14 through 17.) ▲

16.8 Example Let G be the group $D_4 = \{\rho_0, \rho_1, \rho_2, \rho_3, \mu_1, \mu_2, \delta_1, \delta_2\}$ of symmetries of the square, described in Example 8.10. In Fig. 16.9 we show the square with vertices 1, 2, 3, 4 as in Fig. 8.11. We also label the sides s_1, s_2, s_3, s_4, the diagonals d_1 and d_2, vertical and horizontal axes m_1 and m_2, the center point C, and midpoints P_i of the sides s_i. Recall that ρ_i corresponds to rotating the square counterclockwise through $\pi i/2$ radians, μ_i



16.9 Figure

16.10 Table

	1	2	3	4	s_1	s_2	s_3	s_4	m_1	m_2	d_1	d_2	C	P_1	P_2	P_3	P_4
ρ_0	1	2	3	4	s_1	s_2	s_3	s_4	m_1	m_2	d_1	d_2	C	P_1	P_2	P_3	P_4
ρ_1	2	3	4	1	s_2	s_3	s_4	s_1	m_2	m_1	d_2	d_1	C	P_2	P_3	P_4	P_1
ρ_2	3	4	1	2	s_3	s_4	s_1	s_2	m_1	m_2	d_1	d_2	C	P_3	P_4	P_1	P_2
ρ_3	4	1	2	3	s_4	s_1	s_2	s_3	m_2	m_1	d_2	d_1	C	P_4	P_1	P_2	P_3
μ_1	2	1	4	3	s_1	s_4	s_3	s_2	m_1	m_2	d_2	d_1	C	P_1	P_4	P_3	P_2
μ_2	4	3	2	1	s_3	s_2	s_1	s_4	m_1	m_2	d_2	d_1	C	P_3	P_2	P_1	P_4
δ_1	3	2	1	4	s_2	s_1	s_4	s_3	m_2	m_1	d_1	d_2	C	P_2	P_1	P_4	P_3
δ_2	1	4	3	2	s_4	s_3	s_2	s_1	m_2	m_1	d_1	d_2	C	P_4	P_3	P_2	P_1

corresponds to flipping on the axis m_i, and δ_i to flipping on the diagonal d_i. We let

$$X = \{1, 2, 3, 4, s_1, s_2, s_3, s_4, m_1, m_2, d_1, d_2, C, P_1, P_2, P_3, P_4\}.$$

Then X can be regarded as a D_4-set in a natural way. Table 16.10 describes completely the action of D_4 on X and is given to provide geometric illustrations of ideas to be introduced. We should be sure that we understand how this table is formed before continuing. ▲

Isotropy Subgroups

Let X be a G-set. Let $x \in X$ and $g \in G$. It will be important to know when $gx = x$. We let

$$X_g = \{x \in X \mid gx = x\} \qquad \text{and} \qquad G_x = \{g \in G \mid gx = x\}.$$

16.11 Example For the D_4-set X in Example 16.8, we have

$$X_{\rho_0} = X, \qquad X_{\rho_1} = \{C\}, \qquad X_{\mu_1} = \{s_1, s_3, m_1, m_2, C, P_1, P_3\}$$

Also, with $G = D_4$,

$$G_1 = \{\rho_0, \delta_2\}, \qquad G_{s_3} = \{\rho_0, \mu_1\}, \qquad G_{d_1} = \{\rho_0, \rho_2, \delta_1, \delta_2\}.$$

We leave the computation of the other X_σ and G_x to Exercises 1 and 2. ▲

Note that the subsets G_x given in the preceding example were, in each case, subgroups of G. This is true in general.

16.12 Theorem Let X be a G-set. Then G_x is a subgroup of G for each $x \in X$.

Proof Let $x \in X$ and let $g_1, g_2 \in G_x$. Then $g_1x = x$ and $g_2x = x$. Consequently, $(g_1g_2)x = g_1(g_2x) = g_1x = x$, so $g_1g_2 \in G_x$, and G_x is closed under the induced operation of G. Of course $ex = x$, so $e \in G_x$. If $g \in G_x$, then $gx = x$, so $x = ex = (g^{-1}g)x = g^{-1}(gx) = g^{-1}x$, and consequently $g^{-1} \in G_x$. Thus G_x is a subgroup of G. ◆

16.13 Definition Let X be a G-set and let $x \in X$. The subgroup G_x is the **isotropy subgroup of** x. ■

Orbits

For the D_4-set X of Example 16.8 with action table in Table 16.10, the elements in the subset $\{1, 2, 3, 4\}$ are carried into elements of this same subset under action by D_4. Furthermore, each of the elements 1, 2, 3, and 4 is carried into all the other elements of the subset by the various elements of D_4. We proceed to show that every G-set X can be partitioned into subsets of this type.

16.14 Theorem Let X be a G-set. For $x_1, x_2 \in X$, let $x_1 \sim x_2$ if and only if there exists $g \in G$ such that $gx_1 = x_2$. Then $\sim$ is an equivalence relation on X.

Proof For each $x \in X$, we have $ex = x$, so $x \sim x$ and $\sim$ is reflexive.

Suppose $x_1 \sim x_2$, so $gx_1 = x_2$ for some $g \in G$. Then $g^{-1}x_2 = g^{-1}(gx_1) = (g^{-1}g)x_1 = ex_1 = x_1$, so $x_2 \sim x_1$, and $\sim$ is symmetric.

Finally, if $x_1 \sim x_2$ and $x_2 \sim x_3$, then $g_1x_1 = x_2$ and $g_2x_2 = x_3$ for some $g_1, g_2 \in G$. Then $(g_2g_1)x_1 = g_2(g_1x_1) = g_2x_2 = x_3$, so $x_1 \sim x_3$ and $\sim$ is transitive. ◆

16.15 Definition Let X be a G-set. Each cell in the partition of the equivalence relation described in Theorem 16.14 is an **orbit in X under G**. If $x \in X$, the cell containing x is the **orbit of x**. We let this cell be Gx. ■

The relationship between the orbits in X and the group structure of G lies at the heart of the applications that appear in Section 17. The following theorem gives this relationship. Recall that for a set X, we use $|X|$ for the number of elements in X, and $(G : H)$ is the index of a subgroup H in a group G.

16.16 Theorem Let X be a G-set and let $x \in X$. Then $|Gx| = (G : G_x)$. If $|G|$ is finite, then $|Gx|$ is a divisor of $|G|$.

Proof We define a one-to-one map ψ from Gx onto the collection of left cosets of G_x in G. Let $x_1 \in Gx$. Then there exists $g_1 \in G$ such that $g_1x = x_1$. We define $\psi(x_1)$ to be the left coset g_1G_x of G_x. We must show that this map ψ is well defined, independent of the choice of $g_1 \in G$ such that $g_1x = x_1$. Suppose also that $g_1'x = x_1$. Then, $g_1x = g_1'x$, so $g_1^{-1}(g_1x) = g_1^{-1}(g_1'x)$, from which we deduce $x = (g_1^{-1}g_1')x$. Therefore $g_1^{-1}g_1' \in G_x$, so $g_1' \in g_1G_x$, and $g_1G_x = g_1'G_x$. Thus the map ψ is well defined.

To show the map ψ is one to one, suppose $x_1, x_2 \in Gx$, and $\psi(x_1) = \psi(x_2)$. Then there exist $g_1, g_2 \in G$ such that $x_1 = g_1x$, $x_2 = g_2x$, and $g_2 \in g_1G_x$. Then $g_2 = g_1g$ for some $g \in G_x$, so $x_2 = g_2x = g_1(gx) = g_1x = x_1$. Thus ψ is one to one.

Finally, we show that each left coset of G_x in G is of the form $\psi(x_1)$ for some $x_1 \in Gx$. Let g_1G_x be a left coset. Then if $g_1x = x_1$, we have $g_1G_x = \psi(x_1)$. Thus ψ maps Gx one to one onto the collection of right cosets so $|Gx| = (G : G_x)$.

If $|G|$ is finite, then the equation $|G| = |G_x|(G : G_x)$ shows that $|Gx| = (G : G_x)$ is a divisor of $|G|$. ◆

16.17 Example Let X be the D_4-set in Example 16.8, with action table given by Table 16.10. With $G = D_4$, we have $G1 = \{1, 2, 3, 4\}$ and $G_1 = \{\rho_0, \delta_2\}$. Since $|G| = 8$, we have $|G1| = (G : G_1) = 4$. ▲

We should remember not only the cardinality equation in Theorem 16.16 but also that the *elements of G carrying x into g_1x are precisely the elements of the left coset g_1G_x*. Namely, if $g \in G_x$, then $(g_1g)x = g_1(gx) = g_1x$. On the other hand, if $g_2x = g_1x$, then $g_1^{-1}(g_2x) = x$ so $(g_1^{-1}g_2)x = x$. Thus $g_1^{-1}g_2 \in G_x$ so $g_2 \in g_1G_x$.

EXERCISES 16

Computations

In Exercises 1 through 3, let

$$X = \{1, 2, 3, 4, s_1, s_2, s_3, s_4, m_1, m_2, d_1, d_2, C, P_1, P_2, P_3, P_4\}$$

be the D_4-set of Example 16.8 with action table in Table 16.10. Find the following, where $G = D_4$.

1. The fixed sets X_σ for each $\sigma \in D_4$, that is, $X_{\rho_0}, X_{\rho_1}, \cdots, X_{\delta_2}$
2. The isotropy subgroups G_x for each $x \in X$, that is, $G_1, G_2, \cdots, G_{P_3}, G_{P_4}$
3. The orbits in X under D_4

Concepts

In Exercises 4 and 5, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

4. A group G *acts faithfully* on X if and only if $gx = x$ implies that $g = e$.
5. A group G is *transitive* on a G-set X if and only if, for some $g \in G$, gx can be every other x.
6. Let X be a G-set and let $S \subseteq X$. If $Gs \subseteq S$ for all $s \in S$, then S is a **sub-G-set.** Characterize a sub-G-set of a G-set X in terms of orbits in X and G.
7. Characterize a transitive G-set in terms of its orbits.
8. Mark each of the following true or false.

   _____ **a.** Every G-set is also a group.
   _____ **b.** Each element of a G-set is left fixed by the identity of G.
   _____ **c.** If every element of a G-set is left fixed by the same element g of G, then g must be the identity e.
   _____ **d.** Let X be a G-set with $x_1, x_2 \in X$ and $g \in G$. If $gx_1 = gx_2$, then $x_1 = x_2$.
   _____ **e.** Let X be a G-set with $x \in X$ and $g_1, g_2 \in G$. If $g_1x = g_2x$, then $g_1 = g_2$.
   _____ **f.** Each orbit of a G-set X is a transitive sub-G-set.
   _____ **g.** Let X be a G-set and let $H \leq G$. Then X can be regarded in a natural way as an H-set.
   _____ **h.** With reference to (g), the orbits in X under H are the same as the orbits in X under G.
   _____ **i.** If X is a G-set, then each element of G acts as a permutation of X.
   _____ **j.** Let X be a G-set and let $x \in X$. If G is finite, then $|G| = |Gx| \cdot |G_x|$.

9. Let X and Y be G-sets with the *same* group G. An **isomorphism** between G-sets X and Y is a map $\phi : X \to Y$ that is one to one, onto Y, and satisfies $g\phi(x) = \phi(gx)$ for all $x \in X$ and $g \in G$. Two G-sets are **isomorphic** if such an isomorphism between them exists. Let X be the D_4-set of Example 16.8.

a. Find two distinct orbits of X that are isomorphic sub-D_4-sets.

b. Show that the orbits $\{1, 2, 3, 4\}$ and $\{s_1, s_2, s_3, s_4\}$ are not isomorphic sub-D_4-sets. [*Hint:* Find an element of G that acts in an essentially different fashion on the two orbits.]

c. Are the orbits you gave for your answer to part (a) the only two different isomorphic sub-D_4-sets of X?

10. Let X be the D_4-set in Example 16.8.

a. Does D_4 act faithfully on X?

b. Find all orbits in X on which D_4 acts faithfully as a sub-D_4-set.

Theory

11. Let X be a G-set. Show that G acts faithfully on X if and only if no two distinct elements of G have the same action on each element of X.

12. Let X be a G-set and let $Y \subseteq X$. Let $G_Y = \{g \in G \mid gy = y \text{ for all } y \in Y\}$. Show G_Y is a subgroup of G, generalizing Theorem 16.12.

13. Let G be the additive group of real numbers. Let the action of $\theta \in G$ on the real plane $\mathbb{R}^2$ be given by rotating the plane counterclockwise about the origin through θ radians. Let P be a point other than the origin in the plane.

a. Show $\mathbb{R}^2$ is a G-set.

b. Describe geometrically the orbit containing P.

c. Find the group G_P.

Exercises 14 through 17 show how all possible G-sets, up to isomorphism (see Exercise 9), can be formed from the group G.

14. Let $\{X_i \mid i \in I\}$ be a disjoint collection of sets, so $X_i \cap X_j = \varnothing$ for $i \neq j$. Let each X_i be a G-set for the same group G.

a. Show that $\bigcup_{i \in I} X_i$ can be viewed in a natural way as a G-set, the **union** of the G-sets X_i.

b. Show that every G-set X is the union of its orbits.

15. Let X be a transitive G-set, and let $x_0 \in X$. Show that X is isomorphic (see Exercise 9) to the G-set L of all left cosets of G_{x_0}, described in Example 16.7. [*Hint:* For $x \in X$, suppose $x = gx_0$, and define $\phi : X \to L$ by $\phi(x) = gG_{x_0}$. Be sure to show ϕ is well defined!]

16. Let X_i for $i \in I$ be G-sets for the same group G, and suppose the sets X_i are not necessarily disjoint. Let $X_i' = \{(x, i) \mid x \in X_i\}$ for each $i \in I$. Then the sets X_i' are disjoint, and each can still be regarded as a G-set in an obvious way. (The elements of X_i have simply been tagged by i to distinguish them from the elements of X_j for $i \neq j$.) The G-set $\bigcup_{i \in I} X_i'$ is the **disjoint union** of the G-sets X_i. Using Exercises 14 and 15, show that every G-set is isomorphic to a disjoint union of left coset G-sets, as described in Example 16.7.

17. The preceding exercises show that every G-set X is isomorphic to a disjoint union of left coset G-sets. The question then arises whether left coset G-sets of distinct subgroups H and K of G can themselves be isomorphic. Note that the map defined in the hint of Exercise 15 depends on the choice of x_0 as "base point." If x_0 is replaced by g_0x_0 and if $G_{x_0} \neq G_{g_0x_0}$, then the collections L_H of left cosets of $H = G_{x_0}$ and L_K of left cosets of $K = G_{g_0x_0}$ form distinct G-sets that must be isomorphic, since both L_H and L_K are isomorphic to X.

a. Let X be a transitive G-set and let $x_0 \in X$ and $g_0 \in G$. If $H = G_{x_0}$ describe $K = G_{g_0x_0}$ in terms of H and g_0.

b. Based on part (a), conjecture conditions on subgroups H and K of G such that the left coset G-sets of H and K are isomorphic.

c. Prove your conjecture in part (b).

18. Up to isomorphism, how many transitive $\mathbb{Z}_4$ sets X are there? (Use the preceding exercises.) Give an example of each isomorphism type, listing an action table of each as in Table 16.10. Take lowercase names a, b, c, and so on for the elements in the set X.

19. Repeat Exercise 18 for the group $\mathbb{Z}_6$.

20. Repeat Exercise 18 for the group S_3. List the elements of S_3 in the order ι, (1, 2, 3), (1, 3, 2), (2, 3), (1, 3), (1, 2).

SECTION 17 †APPLICATIONS OF G-SETS TO COUNTING

This section presents an application of our work with G-sets to counting. Suppose, for example, we wish to count how many distinguishable ways the six faces of a cube can be marked with from one to six dots to form a die. The standard die is marked so that when placed on a table with the 1 on the bottom and the 2 toward the front, the 6 is on top, the 3 on the left, the 4 on the right, and the 5 on the back. Of course, other ways of marking the cube to give a distinguishably different die are possible.

Let us distinguish between the faces of the cube for the moment and call them the bottom, top, left, right, front, and back. Then the bottom can have any one of six marks from one dot to six dots, the top any one of the five remaining marks, and so on. There are $6! = 720$ ways the cube faces can be marked in all. Some markings yield the same die as others, in the sense that one marking can be carried into another by a rotation of the marked cube. For example, if the standard die described above is rotated 90° counterclockwise as we look down on it, then 3 will be on the front face rather than 2, but it is the same die.

There are 24 possible positions of a cube on a table, for any one of six faces can be placed down, and then any one of four to the front, giving $6 \cdot 4 = 24$ possible positions. Any position can be achieved from any other by a rotation of the die. These rotations form a group G, which is isomorphic to a subgroup of S_8 (see Exercise 45 of Section 8). We let X be the 720 possible ways of marking the cube and let G act on X by rotation of the cube. We consider two markings to give the same die if one can be carried into the other under action by an element of G, that is, by rotating the cube. In other words, we consider each *orbit* in X under G to correspond to a single die, and different orbits to give different dice. The determination of the number of distinguishable dice thus leads to the question of determining the number of orbits under G in a G-set X.

The following theorem gives a tool for determining the number of orbits in a G-set X under G. Recall that for each $g \in G$ we let X_g be the set of elements of X left fixed by g, so that $X_g = \{x \in X \mid gx = x\}$. Recall also that for each $x \in X$, we let $G_x = \{g \in G \mid gx = x\}$, and Gx is the orbit of x under G.

17.1 Theorem **(Burnside's Formula)** Let G be a finite group and X a finite G-set. If r is the number of orbits in X under G, then

$$r \cdot |G| = \sum_{g \in G} |X_g|. \tag{1}$$

† This section is not used in the remainder of the text.

Proof We consider all pairs (g, x) where $gx = x$, and let N be the number of such pairs. For each $g \in G$ there are $|X_g|$ pairs having g as first member. Thus,

$$N = \sum_{g \in G} |X_g|. \tag{2}$$

On the other hand, for each $x \in X$ there are $|G_x|$ pairs having x as second member. Thus we also have

$$N = \sum_{x \in X} |G_x|.$$

By Theorem 16.16 we have $|Gx| = (G : G_x)$. But we know that $(G : G_x) = |G|/|G_x|$, so we obtain $|G_x| = |G|/|Gx|$. Then

$$N = \sum_{x \in X} \frac{|G|}{|Gx|} = |G| \left(\sum_{x \in X} \frac{1}{|Gx|} \right). \tag{3}$$

Now $1/|Gx|$ has the same value for all x in the same orbit, and if we let $\mathcal{O}$ be any orbit, then

$$\sum_{x \in \mathcal{O}} \frac{1}{|Gx|} = \sum_{x \in \mathcal{O}} \frac{1}{|\mathcal{O}|} = 1. \tag{4}$$

Substituting (4) in (3), we obtain

$$N = |G| \text{ (number of orbits in } X \text{ under } G) = |G| \cdot r. \tag{5}$$

Comparison of Eq. 2 and Eq. 5 gives Eq. 1. ◆

17.2 Corollary If G is a finite group and X is a finite G-set, then

$$\text{(number of orbits in } X \text{ under } G) = \frac{1}{|G|} \cdot \sum_{g \in G} |X_g|.$$

Proof The proof of this corollary follows immediately from the preceding theorem. ◆

Let us continue our computation of the number of distinguishable dice as our first example.

17.3 Example We let X be the set of 720 different markings of faces of a cube using from one to six dots. Let G be the group of 24 rotations of the cube as discussed above. We saw that the number of distinguishable dice is the number of orbits in X under G. Now $|G| = 24$. For $g \in G$ where $g \neq e$, we have $|X_g| = 0$, because any rotation other than the identity element changes any one of the 720 markings into a different one. However, $|X_e| = 720$ since the identity element leaves all 720 markings fixed. Then by Corollary 17.2,

$$\text{(number of orbits)} = \frac{1}{24} \cdot 720 = 30,$$

so there are 30 distinguishable dice. ▲

Of course the number of distinguishable dice could be counted without using the machinery of the preceding corollary, but by using elementary combinatorics as often taught in a freshman finite math course. In marking a cube to make a die, we can, by rotation if necessary, assume the face marked 1 is down. There are five choices for the top (opposite) face. By rotating the die as we look down on it, any one of the remaining four faces could be brought to the front position, so there are no different choices involved for the front face. But with respect to the number on the front face, there are $3 \cdot 2 \cdot 1$ possibilities for the remaining three side faces. Thus there are $5 \cdot 3 \cdot 2 \cdot 1 = 30$ possibilities in all.

The next two examples appear in some finite math texts and are easy to solve by elementary means. We use Corollary 17.2 so that we have more practice thinking in terms of orbits.

17.4 Example How many distinguishable ways can seven people be seated at a round table, where there is no distinguishable "head" to the table? Of course there are 7! ways to assign people to the different chairs. We take X to be the 7! possible assignments. A rotation of people achieved by asking each person to move one place to the right results in the same arrangement. Such a rotation generates a cyclic group G of order 7, which we consider to act on X in the obvious way. Again, only the identity e leaves any arrangement fixed, and it leaves all 7! arrangements fixed. By Corollary 17.2

$$(\text{number of orbits}) = \frac{1}{7} \cdot 7! = 6! = 720. \qquad \blacktriangle$$

17.5 Example How many distinguishable necklaces (with no clasp) can be made using seven different-colored beads of the same size? Unlike the table in Example 17.4, the necklace can be turned over as well as rotated. Thus we consider the full dihedral group D_7 of order $2 \cdot 7 = 14$ as acting on the set X of 7! possibilities. Then the number of distinguishable necklaces is

$$(\text{number of orbits}) = \frac{1}{14} \cdot 7! = 360. \qquad \blacktriangle$$

In using Corollary 17.2, we have to compute $|G|$ and $|X_g|$ for each $g \in G$. In the examples and the exercises, $|G|$ will pose no real problem. Let us give an example where $|X_g|$ is not as trivial to compute as in the preceding examples. We will continue to assume knowledge of very elementary combinatorics.

17.6 Example Let us find the number of distinguishable ways the edges of an equilateral triangle can be painted if four different colors of paint are available, assuming only one color is used on each edge, and the same color may be used on different edges.

Of course there are $4^3 = 64$ ways of painting the edges in all, since each of the three edges may be any one of four colors. We consider X to be the set of these 64 possible painted triangles. The group G acting on X is the group of symmetries of the triangle, which is isomorphic to S_3 and which we consider to be S_3. We use the notation for

elements in S_3 given in Section 8. We need to compute $|X_g|$ for each of the six elements g in S_3.

$\|X_{\rho_0}\| = 64$	Every painted triangle is left fixed by ρ_0.
$\|X_{\rho_1}\| = 4$	To be invariant under ρ_1, all edges must be the same color, and there are 4 possible colors.
$\|X_{\rho_2}\| = 4$	Same reason as for ρ_1.
$\|X_{\mu_1}\| = 16$	The edges that are interchanged must be the same color (4 possibilities) and the other edge may also be any of the colors (times 4 possibilities).
$\|X_{\mu_2}\| = \|X_{\mu_3}\| = 16$	Same reason as for μ_1.

Then

$$\sum_{g \in S_3} |X_g| = 64 + 4 + 4 + 16 + 16 + 16 = 120.$$

Thus

$$\text{(number of orbits)} = \frac{1}{6} \cdot 120 = 20,$$

and there are 20 distinguishable painted triangles. ▲

17.7 Example We repeat Example 17.6 with the assumption that a different color is used on each edge. The number of possible ways of painting the edges is then $4 \cdot 3 \cdot 2 = 24$, and we let X be the set of 24 possible painted triangles. Again, the group acting on X can be considered to be S_3. Since all edges are a different color, we see $|X_{\rho_0}| = 24$ while $|X_g| = 0$ for $g \neq \rho_0$. Thus

$$\text{(number of orbits)} = \frac{1}{6} \cdot 24 = 4,$$

so there are four distinguishable triangles. ▲

EXERCISES 17

Computations

In each of the following exercises use Corollary 17.2 to work the problem, even though the answer might be obtained by more elementary methods.

1. Find the number of orbits in $\{1, 2, 3, 4, 5, 6, 7, 8\}$ under the cyclic subgroup $\langle (1, 3, 5, 6) \rangle$ of S_8.
2. Find the number of orbits in $\{1, 2, 3, 4, 5, 6, 7, 8\}$ under the subgroup of S_8 generated by $(1, 3)$ and $(2, 4, 7)$.
3. Find the number of distinguishable tetrahedral dice that can be made using one, two, three, and four dots on the faces of a regular tetrahedron, rather than a cube.
4. Wooden cubes of the same size are to be painted a different color on each face to make children's blocks. How many distinguishable blocks can be made if eight colors of paint are available?

5. Answer Exercise 4 if colors may be repeated on different faces at will. [*Hint:* The 24 rotations of a cube consist of the identity, 9 that leave a pair of opposite faces invariant, 8 that leave a pair of opposite vertices invariant, and 6 leaving a pair of opposite edges invariant.]
6. Each of the eight corners of a cube is to be tipped with one of four colors, each of which may be used on from one to all eight corners. Find the number of distinguishable markings possible. (See the hint in Exercise 5.)
7. Find the number of distinguishable ways the edges of a square of cardboard can be painted if six colors of paint are available and
   a. no color is used more than once.
   b. the same color can be used on any number of edges.
8. Consider six straight wires of equal lengths with ends soldered together to form edges of a regular tetrahedron. Either a 50-ohm or 100-ohm resistor is to be inserted in the middle of each wire. Assume there are at least six of each type of resistor available. How many essentially different wirings are possible?
9. A rectangular prism 2 ft long with 1-ft square ends is to have each of its six faces painted with one of six possible colors. How many distinguishable painted prisms are possible if
   a. no color is to be repeated on different faces,
   b. each color may be used on any number of faces?